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Completely monotone distributions: Mixing, approximation and estimation of number of species

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ABSTRACT

The problem of species richness estimation using complete monotonicity of the distribution of species abundances is considered. Complete monotonicity is the most natural surrogate for k-monotonicity when k is large. The latter model has been considered in the same estimation problem adopting two different approaches which both necessitate selecting the unknown degree of monotonicity k via some chosen criterion. It is shown that such selection procedures can be avoided by appropriately approximating the true completely monotone distribution by a k_n-monotone one such that k_n grows logarithmically as a function of the sample size n. Furthermore, the proposed estimator of the true total number of species is proved to be asymptotically normal. An extended simulation study indicates that it is quite competitive when compared to other available estimators, and this remains true even when complete monotonicity is not satisfied. It is further illustrated how the method can be applied in practice by using four real datasets.
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1. Introduction

1.1. Species richness and shape-constrained estimation

Nonparametric estimation using only shape constraints on the underlying distribution of the data has attracted a lot of attention from both the theoretical and practical perspectives. Shape-constrained estimation uses only the assumption that the density of the true distribution, with respect to some dominating measure, belongs to some class of functions which is entirely described by a particular shape. Shapes that have been considered in continuous or discrete settings (or both) are monotonicity, convexity, log-concavity, bi-log-concavity, unimodality, s-concavity, k-monotonicity and complete monotonicity. We refer the reader for example to Grenander (1956), Keilson and Gerber (1971), Hampel (1987), Groeneboom et al. (2001), Jankowski and Wellner (2009), Cule et al. (2010), Balabdaoui and Wellner (2007), Durot et al. (2013), Balabdaoui et al. (2013) and Lefèvre and Loisel (2013), Doss and Wellner (2016), Balabdaoui and Jankowski (2016), Chee and Wang (2016), Dümbgen et al. (2017), Giguelay (2017), Giguelay and Huet (2018), Balabdaoui et al. (2018) and Balabdaoui and de Fournas-Labrosse (2020). Shape constrained estimation has many attractive features. For example, one avoids the hurdle of having to find an optimal tuning parameter, as with bandwidth selection within the kernel estimation framework. Estimators, such as the maximum likelihood or least squares estimator can be efficiently computed using fast algorithms such as active set, interior point, iterative convex minorant or support reduction algorithms; see for example Wang (2007), Rufibach (2007) and Groeneboom et al. (2008). Rates of convergence for such estimators, at a point or in some global measure, can be derived for most of the aforementioned shape constraints, although the limit

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distribution may involve elaboration of some sophisticated theoretical arguments. In the discrete setting which we are considering in this paper, the rate of convergence of shape-constrained estimators may attain the parametric rate $\sqrt{n}$ as opposed to slower rates in the continuous case. For a more extended discussion on this point, we refer to Balabdaoui and de Fournas-Labrosse (2020).

In this paper, we are interested in the particular class of completely monotone probability mass functions (pmf) as well as in applying some of the remarkable theoretical properties of its elements to estimate the total number of species in a given population, better known under the term of species richness. To explain the motivation and methodology, we first give a brief review of this problem. Estimation of species richness is a well-known problem in statistics because it can be encountered in various fields such as biology, metagenomics, ecology, entomology, vocabulary richness, to name only a few. A non-exhaustive list of references on nonparametric methods of estimating species richness includes Good (1953), Efron and Thisted (1976), Bunge and Fitzpatrick (1993), Chao and Bunge (2002), Böhning and Kuhnert (2006), Mao and Lindsay (2007), Durot et al. (2015), Chee and Wang (2016) and Giguelay and Huet (2018). In the papers by Zhang and Grabchak (2016) and Zhang et al. (2020), brought to our attention by a referee, the authors consider the problem of species richness using an entropic basis. In this basis, the unknown total number of species can be written as a weighted average where all the weights are equal to 1; see (3) in Zhang et al. (2020). Using an empirical estimator of the elements of the basis (which can be shown to be a uniformly minimum variance unbiased estimator), this representation can be then used to find an estimator of the total number of species. Note that although the elements of this entropic basis are given by sums of geometric pmfs, it is not clear how the approach of Zhang et al. (2020) can be related to the one studied in this article. Another referee pointed us to the interesting work of Orlitsky et al. (2004) and Acharya et al. (2017). The method considered in these works is based on finding the high profile rather than the maximum likelihood estimator. There, the profile is defined to be the set of the numbers of occurrence of the observed species, that is, the set of observed abundances. For a precise definition of this estimator, some of its fundamental properties, and numerical approaches to compute it, we refer to Orlitsky et al. (2004). Their work has motivated that of Anevski et al. (2017), who studied the almost sure consistency of this high profile estimator, called there the pattern maximum likelihood estimator. The book by Zhang (2017) gives a thorough account of the various implications of the Good–Turing's formula; see Good (1953). This formula, which has been the subject of an extensive investigation, is probably the earliest nonparametric estimator of the proportion of unseen species in some given population. In short, this formula computes the fraction of the number of species which are represented by exactly one individual over the total sample size of all the individuals drawn from the population. In essence, estimation of species richness can be really viewed as a statistical extrapolation problem in the following sense: given the observed abundances (the number individuals of a certain species) in a certain population, we would like to "guess" the abundance of the unseen species from the information collected about the species which have very small abundances. This guess should of course depend, in some way, on the choice made about the statistical model for the abundances.

In species richness, mixture models have played a crucial role since they have a great ability to capture possible existing heterogeneity among the species. Poisson mixtures were first used in this problem by Norris and Pollock (1998) and became quite popular since. However, as shown by Wang and Lindsay (2005), the model suffers from different drawbacks such as the so-called "boundary problem". This can occur when the MLE of the mixing distribution in the mixture of Poisson puts mass near 0. This causes overestimation of the unknown number of species. Motivated by circumventing this issue, alternative mixture models were considered. This includes recent work of Durot et al. (2015), Chee and Wang (2016) and Giguelay and Huet (2018). The common point of these works is the use of shape-constrained estimation of the abundances. In Durot et al. (2015), the distribution of abundances is assumed to be convex on the set of non-negative integers $\mathbb{N} = \{0, 1, \ldots\}$. In Giguelay and Huet (2018), convexity is generalized to k-monotonicity on $\mathbb{N}$ (convexity is equivalent to 2-monotonicity). More specifically, for $k \geq 1$ some given integer, a real sequence $c \equiv (c(j))_{j\in\mathbb{N}}$ is said to be k-monotone if it satisfies that $(-1)^r \Delta^r c(j) \geq 0$ for all $r \in \{0, \ldots, k\}$ and $j \in \mathbb{N}$, where $\Delta^r c(j) = \sum_{i=0}^{r} (-1)^{i+r} \binom{r}{i} c(i+j)$ is the rth order forward operator. It was shown in Lefèvre and Loisel (2013) that a probability mass function (pmf) p is k-monotone if and only if we can find a sequence $(\pi_l^k)_{l\in\mathbb{N}}$ of probabilities summing up to 1 such that for all $j \in \mathbb{N}$

$$p(j) = \sum_{l\geq 0} \pi_l^k Q_l^k(j), \tag{1}$$

where

$$Q_l^k(j) = \frac{\binom{l-j+k-1}{k-1}}{\binom{l+k}{l}} \mathbf{1}_{\{0\leq j\leq l\}} \tag{2}$$

is the k-monotone spline with support $\{0, \ldots, l\}$. Thus, k-monotonicity defines a mixture model, with a discrete mixing distribution from which the support endpoints $l \in \mathbb{N}$ are drawn. The mixing probability π_l^k can be given by the useful inverse formula

$$\pi_l^k = (-1)^k \Delta^k p(l) \binom{l+k}{l}, \tag{3}$$

which we shall use in Section 2. Here, the superscript k in π_l^k and Q_l^k corresponds to the degree of monotonicity in the assumed model. In the framework of k-monotonicity as a shape-constrained model for abundances, the estimation

method in both Durot et al. (2015) and Giguelay and Huet (2018) uses a principle which we call in the next section the "one cannot see what cannot be seen" principle. This allows for identifying the probability that some species of the population under consideration are not observed and deducing the estimator for the total number of species from the number of those which can be observed. In the next section we will return to the estimation approach of Giguelay and Huet (2018) in much more detail.

Chee and Wang (2016) have also considered the problem of species richness under k-monotonicity of the true pmf of abundances. As mentioned by the authors in their introduction, this shape-constraint seems to be quite appropriate in the case where a population is dominated by a large number of rare species. Their approach is however different from that of Giguelay and Huet (2018) in two aspects. First, Chee and Wang (2016) consider only a subset of k-monotone distributions, more precisely, only the k-monotone pmfs generated by the discrete k-monotone splines

$$k_\theta(j) = \frac{(\theta - j)_+^{k-1}}{\sum_{j=0}^{\lfloor\theta\rfloor}(\theta - j)^{k-1}},$$

where $\theta \geq 1$ and $\lfloor\cdot\rfloor$ denotes the integer part. As pointed out by the authors, the kernel can be seen as a discretized version of the density function of Beta$(1, k)$ which generates the class of k-monotone Lebesgue densities on $(0, \infty)$ considered for example in Balabdaoui and Wellner (2007) and Balabdaoui and Wellner (2010). Second, Chee and Wang (2016) use the maximum likelihood estimation. If $p(j, F) = \int_0^\infty k_\theta(j)dF(\theta)$ for $j \in \mathbb{N}$ denotes the pmf with F being the unknown mixing distribution, Chee and Wang (2016) consider the unconditional maximum likelihood estimation approach where the goal is to find the pair $(\widehat{N}, \widehat{F})$ maximizing the full multinomial likelihood

$$L(N, F) = \frac{N!}{(N-s)!\prod_{j\in\mathbb{N}^*} n_j!}\, p(0, F)^{N-s} \prod_{j\in\mathbb{N}^*} p(j, F)^{n_j}$$

where n_j is the number of species with abundance $j \in \mathbb{N}^*$, and $s = \sum_{j\in\mathbb{N}^*} n_j$ is the total number of observed species. As indicated by the authors, it follows from the work of Sanathanan (1972) that $L(N, F)$ can be decomposed as

$$\begin{aligned} L(N, F) &= L_m(N, F) \times L_c(F) \\ &= \frac{N!}{(N-s)!s!}\, p(0, F)^{N-s}(1 - p(0, F))^s \\ &\quad \times \frac{s!}{\prod_{j\in\mathbb{N}^*} n_j!} \prod_{j\in\mathbb{N}^*} \left\{ \frac{p(j, F)}{1 - p(0, F)} \right\}^{n_j}. \end{aligned}$$

Hence, one can maximize first the conditional likelihood $L_c(F)$ in F and then use this conditional MLE, $\widehat{F}_c$ say, to obtain the MLE $\widehat{N}_c = \left\lfloor \frac{s}{1-p(0,\widehat{F}_c)} \right\rfloor$, which maximizes $L(N, \widehat{F}_c)$ over $\mathbb{N}$. Interestingly, and as proved by Wang and Lindsay (2005), the MLE $(\widehat{N}_u, \widehat{F}_u)$ obtained by the full likelihood $L(N, F)$ satisfies $\widehat{N}_u = \left\lfloor \frac{s}{1-p(0,\widehat{F}_u)} \right\rfloor$, showing that the conditional and full MLE have similar extrapolation strategies once the mixing distribution has been estimated.

1.2. Motivation, contributions and organization of the paper

In Table 3 of Chee and Wang (2016), the authors present the results of simulations in different mixture settings, including mixture of geometrics with two components. As they mention in their introduction, when $k \to \infty$, the k-monotone model converges to the complete monotone one; i.e., to the set of pmfs p such that $(-1)^k \Delta^k p(i) \geq 0$ for all $i \in \mathbb{N}$ and $k \in \mathbb{N}$. The aforementioned convergence will be elaborated in more detail in the next section. A very important result at the intersection of Analysis and Statistics is the one, proved by Steutel (1969): a pmf p is completely monotone if and only if it is a mixture of geometric pmfs. As explained in Balabdaoui and de Fournas-Labrosse (2020), this result is tightly connected to the well-known Hausdorff Theorem; see Feller (1971). To obtain the species richness estimator, Chee and Wang (2016) fit the best k-monotone MLE by selecting the parameter k using the Akaike-type criterion. Interestingly, Giguelay and Huet (2018) also consider a selection procedure of k. If p_0 is the true pmf, their procedure is based on successively testing

$$H_{0,k} : p_0 \text{ is } k\text{-monotone and not } (k+1)\text{-monotone}$$

versus

$$H_{1,k} : p_0 \text{ is } (k+1)\text{-monotone}$$

for $k \in \{1, \ldots, k_{max}\}$ for some given k_{max} and stopping when rejection of the null hypothesis occurs for the first time. The integer $\widehat{k}$ immediately preceding the first integer leading to the rejection is taken to be the estimator of the true degree of monotonicity. In their Section 2.2.2, Giguelay and Huet (2018) note that when p_0 is k-monotone with large k, their testing procedures have small power. This is expected since the differences between k- and $(k+1)$-monotonicity become harder

to detect as k increases. This finding can be also linked to the main subject of this manuscript: complete monotonicity. In fact, when k is large, p_0 is "almost" completely monotone. Hence, when the distribution is almost at the limit with a large degree of monotonicity k, one would expect that distinguishing between k and $k+1$ becomes hard. In Section 5, this finding is confirmed numerically since our test for complete monotonicity often fails to reject the null hypothesis for different k-monotone distributions with large k.

One of the main contributions of this work is to provide a new way of estimating species richness based on the complete monotonicity constraint. Our approach does not build upon selection of the degree of monotonicity. Instead, the latter is estimated to ensure that the estimator of the probability of not observing a species belonging to the population under consideration is asymptotically normal. The estimator is found to grow logarithmically with the observed number of species. A byproduct of our analysis is a new approximation result which generalizes the well-known Stirling approximation for the binomial coefficient $\binom{2k}{k}$ for large integers k. Although complete monotonicity characterizes all mixtures of geometric pmfs, we shall make the assumption that the distribution of abundances is a finite mixture of such pmfs. We shall discuss why such an assumption is not restrictive, while benefiting from the great convenience it offers for analyzing our estimator of the size of a population. Before our method can be used, a nonparametric test could be first applied to check the validity of complete monotonicity. This test, whose critical region is found via a bootstrap procedure, is based on the completely monotone LSE studied in Balabdaoui and de Fournas-Labrosse (2020). The completely monotone LSE, $\widehat{p}_n$ say, results from the L_2-projection of the empirical estimator onto the class of completely monotone functions defined on $\mathbb{N}$ with a finite L_2-norm. The authors also give a detailed analysis of the asymptotic behavior of $\widehat{p}_n$. In particular, it is shown that this estimator converges at the $\sqrt{n}$ rate in any L_q distance for $q \in [2, \infty]$. The limit distribution of $\widehat{p}_n$ is also established, where it is also proved that the process $\{\sqrt{n}(\widehat{p}_n(x) - p_0(x)),\ x \in \mathbb{N}\}$ converges weakly to some process $\{\widehat{\mathbb{L}}(x), x \in \mathbb{N}\}$, the solution of a well-defined minimization problem. In the simulation study described below, we find that our estimator often outperforms the ones considered by Chee and Wang (2016) and Giguelay and Huet (2018). This high performance comes with no additional computational cost as our estimation procedure relies only on the immediate use of the estimator of the degree of monotonicity, which can be directly computed from the observed data.

The paper is organized as follows. In the next section, we give the precise meaning of convergence of k-monotonicity to complete monotonicity. In Section 3, we present our approach to estimating species richness of a given population under complete monotonicity. Constructing this estimator on the basis of the empirical estimator of the abundances for the observed species, we show in Section 4 that the resulting estimator of the total number of species is asymptotically normal. Assessment of the performance of our method and comparison to the methods of Chee and Wang (2016) and Giguelay and Huet (2018) based on synthetic data are the subject of Section 5. A bootstrap test based on the completely monotone LSE is presented in Section 6, where we provide its level and power under several scenarios. In Section 7, we apply our method to four real datasets, for example the insurance claims and Shakespeare datasets, which were also considered in Chee and Wang (2016). To keep the reading fluid, we deferred the proofs to the Appendix .

2. Complete monotonicity: mixing and approximation

We would like to establish convergence of k-monotonicity to complete monotonicity as $k \to \infty$ in a precise way. Even though intuitive (since the number of inequalities satisfied by the discrete Laplacian grows), such a convergence will explain why one would be tempted to replace the k-monotone by the completely monotone model in case it is believed that the underlying distribution is k-monotone with large k.

2.1. The case of a fixed completely monotone pmf

Equivalence of complete monotonicity of a pmf to being a mixture of geometric pmfs goes back to Steutel (1969), and is tightly connected to the celebrated Hausdorff Theorem; see for example Section 3 of Chapter VII in Feller (1971).

To prepare the ground for the complete monotone approximation to the k-monotone model for $k \to \infty$, consider first a completely monotone pmf p and the associated mixing probability distribution F on $[0, 1]$ such that $F(1-) = 1$ and

$$p(j) = \int_0^1 \alpha^j (1-\alpha) dF(\alpha) \tag{4}$$

for all $j \in \mathbb{N}$. We will now show that the mixing distribution F can be expressed as a limit of a weighted sum of the mixing probabilities appearing in the mixture representation of p given in (1), when p is viewed as a k-monotone pmf. Our proof is based on the important results due to Feller (1971) on moment problems; see Section 3 of Chapter VII in Feller (1971).

Proposition 2.1. *Let p be given by (4). The (unique) mixing probability distribution F is given by*

$$F(\alpha) = \lim_{k\to\infty} \sum_{0\le j\le k\alpha} \binom{k}{j} (-1)^{k-1-j} \Delta^{k-1-j} p(j)$$

for any $\alpha \in [0, 1)$ which is a continuity point of F.

2.2. Convergence of a k-monotone sequence of pmfs

The preceding proposition provides a way of linking the mixing probabilities in the k-monotone decomposition to the mixing distribution of a completely monotone discrete probability. Recall that a completely monotone pmf p, viewed as a k-monotone pmf, can be given by the following mixture spline representation

$$p(i) = \sum_{l \geq 0} \pi_l^k Q_l^k(i),$$

for all $i \in \mathbb{N}$, where the π_l^k can be given by the expression in (3), which we recall here for convenience:

$$\pi_l^k = (-1)^k \Delta^k p(l) \binom{l+k}{l}.$$

This inversion formula can be found for example in (2.16) in Lefèvre and Loisel (2013). Now, replacing k by $k-1-j$ and setting $l = j$ in π_l^k the mixing distribution F corresponding to p can also be given by

$$F(\alpha) = \lim_{k \to \infty} \sum_{0 \leq j \leq k\alpha} \pi_j^{k-1-j} \frac{k}{k-j}.$$

Therefore, F appears to be the limit of a special weighted combination of the probabilities π_l^k, $k \in \mathbb{N}$.

Now, suppose that we have $(p^k)_{k \in \mathbb{N}}$, a family of probability mass functions, such that p^k is known to be k-monotone for each $k \in \mathbb{N}$. The question we would like to answer is whether there exists a sufficient condition under which p^k converges pointwise to a completely monotone p. The answer to this question turns out to be positive and is furthermore related to the previous problem where we had a fixed completely monotone pmf. We have the following result.

Proposition 2.2. *For $\alpha \in [0, 1)$, define the non-decreasing function*

$$F^k(\alpha) = \sum_{0 \leq j \leq k\alpha} \pi_j^{k-1-j} \frac{k}{k-j}$$

for $\alpha \in [0, 1)$. If F^k converges weakly to some distribution function F on $[0, 1)$ such that $F(1-) = 1$, then the sequence $(p^k)_{k \in \mathbb{N}}$ converges to the completely monotone pmf p given by $p(i) = \int_0^1 \alpha^i (1-\alpha) dF(\alpha)$.

As already mentioned above, if it is expected that k-monotonicity leads to complete monotonicity as k grows, such a statement merely refers to the fact that as $k \to \infty$ the constraints $(-1)^r \Delta^r p(i) \geq 0$ will be fulfilled for all $i \in \mathbb{N}$ (since the set $\{0, \ldots, k\}$ converges in the sense of the union to $\mathbb{N}$). However, it is certainly clear whether p is already a completely monotone pmf, in which case the statement is obvious. Proposition 2.2 formalizes what convergence of k-monotonicity to complete or ∞-monotonicity actually means.

3. Species richness estimation under complete monotonicity

3.1. A motivating example

The following real data example motivates our approach based on the completely monotone model. The dataset comprises the number of occurrences of the words that Shakespeare used in his writings. The underlying question is: how many words did he actually know? The question has preoccupied a number of statisticians; see e.g. Efron and Thisted (1976). Here, the species are the words and N we want to estimate is the total number of all words Shakespeare knew, including the ones which he did not use or were not observed for some unknown reason. The dataset of words used by Shakespeare can be found in Spevack (1968). Chee and Wang (2016) noticed a discrepancy between the counts used by Efron and Thisted (1976) and those given in Spevack (1968). Then, they recomputed the word counts up to 100 occurrences directly from the Appendix provided in Spevack (1968); see Table 5 in Chee and Wang (2016). In Fig. 1 are plotted the empirical estimator of abundances in solid line and the complete monotone LSE studied by Balabdaoui and de Fournas-Labrosse (2020) in bullets. The two estimators are very close, which suggests that the complete monotonicity assumption is appropriate for this dataset. This is also confirmed by our bootstrap test for complete monotonicity, which we describe below. The same dataset will be considered again in Section 5. In that section, we will consider other datasets for which complete monotonicity seems to be a very appropriate shape constraint for the abundances of the species. As for k-monotonicity, complete monotonicity is a suitable model whenever there is a large number of rare species. This feature, which has been noted by Chee and Wang (2016), can be observed for example in the case of the Shakespeare data, where over 60% of the words used by the writer occurred only once or twice.

3.2. General assumptions and notation

Recall that the problem in which we are interested is that of estimating species richness of some population. We give again the description of the problem, the nature of the data involved and the assumptions that we will make to achieve

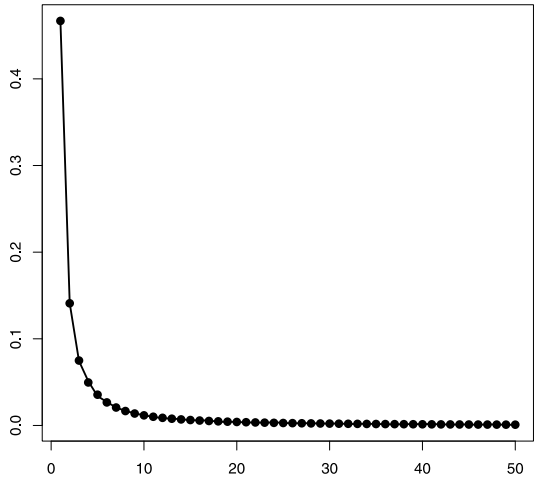

Fig. 1. In solid line is the empirical estimator of the frequencies of words used by Shakespeare. In bullets is the complete monotone LSE.

this goal. With some sampling mechanism, we draw individuals at random from a given population with an unknown total number of distinct species N. To give a simple illustration, imagine a zoologist goes to some jungle with the aim of estimating the total number of its species. During their journey, the zoologist could only observe the following 50 animals: 20 lions, 20 elephants, 8 giraffes and 2 zebras, while there are also tigers in the same area. In this example, there are 5 species in total but only 4 could be observed. The main task of the zoologist is to use the information they have about the abundances of the 4 observed species to infer about the missed species in the sample. The resulting abundances of any sampling mechanism which allows for random selection of individuals from the population depends of course on the actual (but unknown) proportions of the species in this population. If it is clear that the observed abundances admit a joint multinomial distribution whose probability vector is furnished by these propositions, a simple use of this likelihood does not allow us to estimate the probability of not seeing some species. Instead of working with this likelihood, it is more advantageous to use the distribution of the abundances. In this alternative way of modeling the problem, the proportions of the species in the population act as a latent layer, and hence the distribution of the abundances should be a mixture thereof.

More formally, let N be the true total number of species in some population. For each species $i = 1, \dots, N$ let us denote X_i the corresponding abundance, that is the number of observed individuals belonging to species i. Following Sanathanan (1972), Durot et al. (2013, 2015), Chee and Wang (2016) and Giguelay and Huet (2018) we assume that

Assumption A1. $X_1, \dots, X_N$ are i.i.d. random variables generated by an unknown distribution with pmf p_0.

Note that p_0 is defined on the whole set of integers. Since not all species can be observed, we only observe those X_i's satisfying necessarily $X_i \geq 1$. As the indices given to the different species are interchangeable, we can assume that we observe $X_1, \dots, X_S$, where S is the number of species with strictly positive abundances. From Assumption A1 it follows that $S \sim \text{Bin}(N, 1 - p_0(0))$, and conditionally on S, $X_1, \dots, X_S$ are i.i.d. from the truncated pmf

$$p_0^+(j) = \frac{p_0(j)}{1 - p_0(0)} \mathbb{1}_{j \geq 1}. \tag{5}$$

Estimation of N, the parameter of interest, and $p_0(0)$ are tightly connected. Indeed, if $p_0(0)$ were known, then the fact that $S \sim \text{Bin}(N, 1 - p_0(0))$ gives the consistent estimator of N

$$\widehat{N} = \left\lfloor \frac{S}{1 - p_0(0)} \right\rfloor$$

where consistency means that $\widehat{N}/N \to 1$ in probability as $N \to \infty$. The value $p_0(0)$ is usually unknown, and hence needs to be estimated. The idea is to link this value in some identifiable way to the truncated distribution p_0^+. Under the assumption that p_0 is k-monotone, Giguelay (2017) proposed an estimator of $p_0(0)$ which is built upon the following principle, introduced by Durot et al. (2015) to ensure identifiability of $p_0(0)$.

The "One cannot see what cannot be seen" principle: Let p_0 be the true pmf of species abundances in some given population. Then, p_0 is k-monotone if and only if there exists a sequence of mixing probabilities $(\pi_l^k)_{l \in \mathbb{N}}$ summing up to 1 such that

$$p_0(j) = \sum_{l \geq 0} \pi_l^k Q_l^k(j),$$

for all $j \in \mathbb{N}$, where we recall that Q_l^k is the k-monotone spline with support $\{0, \ldots, l\}$ given in (2). Since Q_0^k is the Dirac distribution at 0 (hence gives total mass 1 to zero abundances), this component should not contribute to the mixture, yielding $\pi_0^k = 0$. This equality can be then used to identify the probability $1 - p_0(0)$ in a unique way. Indeed, using the inversion formula in (3), the condition $\pi_0^k = 0$ can be equivalently replaced by $\Delta^k p_0(0) = 0$, that is

$$p_0(0) + \sum_{j=1}^{k} (-1)^j \binom{k}{j} p_0(j) = 0. \tag{6}$$

or equivalently

$$\frac{1}{1 - p_0(0)} = 1 - \sum_{j=1}^{k} (-1)^j \binom{k}{j} p_0^+(j). \tag{7}$$

We would like to stress the important fact that the identity in (7) is not general for all k-monotone pmfs but only valid for the sub-class of pmfs of abundances.

Based on the observed abundances $X_1, \ldots, X_S$, let $\widehat{p}^+$ be a consistent estimator of the true pmf p_0^+. Then, an estimator of N can be expressed as

$$\widehat{N}^k = \left\lfloor S\left(1 - \sum_{j=1}^{k} (-1)^j \binom{k}{j} \widehat{p}^+(j)\right) \right\rfloor,$$

where the subscript k reminds us that the estimation is done under k-monotonicity. An important fact related to shape-constrained classes is that the truncated pmf p_0^+ satisfies the same shape as p_0 on the support $\{1, 2, \ldots\}$. Thus, under k-monotonicity, a good estimator $\widehat{p}^+$ can be taken for example to be the k-monotone LSE studied by Giguelay (2017), which is known to converge at the $\sqrt{n}$-rate. Another possible estimator is the empirical estimator. The advantage of choosing the latter is that it enables one to derive asymptotic normality of $\widehat{N}^k$ as $N \to \infty$; see Section 3.1 in Durot et al. (2013) and Section 5 in Giguelay and Huet (2018). For $k = 2$, asymptotic normality for the estimator based on the convex (2-monotone) LSE can be derived under some additional assumptions such as boundedness of the support of the true pmf; see Section 3.1 in Durot et al. (2013). Generalization of such a result is not yet available for the k-monotone LSE.

3.3. Specific assumptions - complete monotonicity

Under the k-monotonicity assumption of the true pmf of abundances, what value of k should be taken? Giguelay and Huet (2018) propose a selection procedure based on testing the null hypothesis that the true pmf of abundances is k-monotone and not $(k+1)$-monotone against the alternative that it is $(k+1)$-monotone for $k \in \{1, \ldots, k_{\max}\}$, for some chosen $k_{\max}$. The selected k, $\widehat{k}$ say, is the first integer which is immediately preceding the first $k \in \{1, \ldots, k_{\max}\}$ leading to rejection of the null hypothesis. The test statistics and their asymptotic properties are described in Section 2.1 in Giguelay and Huet (2018). The consistency result for $\widehat{k}$ is established in Theorem 3 in Giguelay and Huet (2018). However, and as observed by these authors, the selection procedure tends to overestimate the true k when the latter is large. This overestimation can be linked to the low power of the k-monotonicity test procedures in this case. Indeed, in Giguelay and Huet (2018) it is shown that when k is large, the testing procedure for k-monotonicity that they have constructed has low power for rejecting the null hypothesis, that is, rejecting that p_0 is k- but not $(k+1)$- monotone. This means that the tests have difficulty to distinguish between k and $k+1$, which explains the aforementioned overestimation. Note that since $\widehat{k}$ is random, the identity in (7) is not exact, which makes derivation of the asymptotic distribution of $\widehat{N}^{\widehat{k}}$ more complicated.

Based on the results in Section 2, when k is large, k-monotonicity can be approximated by complete monotonicity. This leads us to our second assumption.

Assumption A2. the true pmf p_0 is completely monotone on $\mathbb{N}$, that is, a mixture of geometric pmf's.

The goal now is to construct an estimator of N under this model, where the selection procedure is replaced by a judicious choice of k which can make inference possible for N large. As it will be described in more detail below, our procedure does not aim at selecting the best k but at finding the parameter $k = k_S$, where S is the number of observed species, which could be viewed as an adequate representative of complete monotonicity. In Section 4, this fitted k will be denoted k_n where n should be understood as the realization of S. The use of the customary subscript "n" in that section has the goal of keeping the reading more fluid. Note that complete monotonicity actually means that $k = \infty$, and hence talking about selection makes only little sense in this framework. One of the practical features of our approach is that it works even when complete monotonicity is not satisfied: it is enough that k-monotonicity holds for some large k (but not necessarily very large). In this case, our estimator has a much smaller variance compared to $\widehat{N}^k$ where k is the true parameter. Indeed, it can be shown that the variance of $\widehat{N}^k$ can reach very large values, making any reasonable inference impossible. Theorem 4.2 shows that our estimator grows logarithmically with the number of species. This logarithmic

order is dictated by the order of variance as a function of k and the Lindeberg–Feller Central Limit Theorem condition. Consequently, it can be shown that our estimator of species richness is asymptotically normal.

To make our task more feasible, we make the following assumption.

Assumption A3. the true pmf p_0 is a finite mixture of geometric pmfs with number of components $m \geq 2$.

Some remarks are in order. Assumption A3 does not specify the number of components of the finite mixture. Hence, the number of these components are left totally unknown, which makes the model used semi-parametric. Our approach relies on finiteness of the mixture and our line of proof cannot be straightforwardly adapted to general mixtures. However, assuming that the mixture is finite is not very restrictive for the following reasons. First of all, any mixture of geometric pmfs can be approximated by a finite mixture, as a consequence of strong consistency of the empirical cumulative distribution function (cdf) to the true one. Here, by the true cdf we mean of course the mixing distribution function as in (4). Second of all, nonparametric estimators such the LSE or MLE are necessarily finite mixtures of geometric pmfs. In fact, for the LSE this follows from Proposition 2.3 of Balabdaoui and de Fournas-Labrosse (2020), whereas the same holds for the MLE as an immediate consequence of Theorem 3.1 due to Lindsay (1983). This means that the most important information can be captured by fitting a finite mixture to the data.

The starting point for constructing our estimator of the unknown N is the application of the "one cannot see what cannot be seen principle" in its limit form. Under Assumption A3, there exist an integer $m \geq 1$, probabilities $\pi_1, \ldots, \pi_m$ with $\sum_{j=1}^m \pi_j = 1$ and $0 \leq \alpha_1 < \cdots < \alpha_m < 1$ such that the true pmf of abundances p_0 can be given by

$$p_0(j) = \pi_1 \alpha_1^j (1-\alpha_1) + \cdots + \pi_m \alpha_m^j (1-\alpha_m),$$

for $j \in \mathbb{N}$. The way this mixture model can be interpreted is that the species are assigned to m different sub-groups with respective probabilities $\pi_1, \ldots, \pi_m$. Now, let assume that $\alpha_1 = 0$. Then, the first pmf in this mixture is equal to the Dirac distribution at 0. Using the above representation, this means that with probability π_1, a species belongs to the sub-group which never shows up in the sample. Since p_0 is the pmf of the distribution of abundances, this could only make sense if $\pi_1 = 0$. In other words, the distribution p_0 can only be described by components which still allow for observing some individuals, and this means that it is impossible under this scheme to have the Dirac distribution as an active component in the mixture model. This is what is exactly referred to as the "one cannot see what cannot be seen" principle: The species which can never be observed have 0 contribution to the distribution of abundances.

From the identity $(-1)^k \Delta^k p_0(j) = \sum_{l=0}^k (-1)^l \binom{k}{l} p_0(j+l)$, it follows that

$$\begin{aligned}(-1)^k \Delta^k p_0(0) &= \sum_{l=0}^k (-1)^l \binom{k}{l} p_0(l) \\ &= \pi_1 (1-\alpha_1)^{k+1} + \cdots + \pi_m (1-\alpha_m)^{k+1}.\end{aligned} \tag{8}$$

Thus, if $\alpha_1 = 0$, the application of the "one cannot see what cannot be seen principle" implies that

$$\lim_{k\to\infty} (-1)^k \Delta^k p_0(0) = \pi_1 = 0. \tag{9}$$

The preceding identity is going to be crucial in establishing asymptotic normality of our estimator defined in Section 4. Note that if p_0 is a general mixture of geometric pmfs, then the "one cannot see what cannot be seen"principle implies that $F_0(0) = 0$, with F_0 the mixing distribution corresponding to p_0. Thus, 0 is a continuity point of F_0 and by Proposition 2.1, we must have

$$0 = F_0(0) = \lim_{k\to\infty} \pi_0^{k-1}$$

where π_0^{k-1} is the mixing probability corresponding to δ_0. But this also implies that $\lim_{k\to\infty} \pi_0^k = 0$ and again by the relationship in (3), we get that the identity in (9) should hold.

In the next section, we define our estimator and show its asymptotic normality, which will follow from Theorems 4.2 and 4.3. One of the key results to building our arguments is formulated in the auxiliary Theorem 4.1.

4. Asymptotic normality of the species richness estimator under complete monotonicity

Recall that in species richness estimation, the sample size is equal to the number of species that were observed. In fact, the data consist of $X_1, \ldots, X_S$, the abundances corresponding to the observed species. Since the true pmf p_0 is assumed to be a finite mixture of geometric pmfs (Assumption A3), it follows that conditionally on S, $X_1, \ldots, X_S$ are i.i.d. from the truncated pmf $p_0^+ = p_0/(1-p_0(0))$, which is also a finite mixture of geometric distributions, but now with support $\mathbb{N} \setminus \{0\} = \{1, 2, \ldots\}$. If $0 < \alpha_1 < \cdots < \alpha_m < 1$ are the failure probabilities and $\pi_1, \ldots, \pi_m$ the corresponding mixing probabilities in the mixture representation of p_0, then

$$p_0^+(j) = \tilde{\pi}_1 (1-\alpha_1) \alpha_1^{j-1} + \cdots + \tilde{\pi}_m (1-\alpha_m) \alpha_m^{j-1} \tag{10}$$

for $j \in \mathbb{N} \setminus \{0\}$, where

$$\tilde{\pi}_i = \frac{\alpha_i \pi_i}{\alpha_1 \pi_1 + \cdots + \alpha_m \pi_m}$$

for $i = 1, \ldots, m$. For ease of exposition, we shall adopt in this sequel the usual notation n. This means that we will replace any occurrence of S by n. Thus, n should be viewed as a random realization from the distribution of number of the seen species.

4.1. Intermediate results

For some $k_n \geq 1$ to be determined later, and $X_1, \ldots, X_n$ i.i.d. $\sim p_0^+$, define the independent variables Z_{ni} and Y_{ni} for $i = 1, \ldots, n$ as

$$Z_{ni} = \frac{1}{\sqrt{n}} \sum_{l=1}^{k_n} (-1)^l \binom{k_n}{l} \left(\mathbb{1}_{X_i = l} - p_0^+(l) \right)$$

and

$$Y_{ni} = \sqrt{n} Z_{ni} = \sum_{l=1}^{k_n} (-1)^l \binom{k_n}{l} \left(\mathbb{1}_{X_i = l} - p_0^+(l) \right).$$

The random variable Y_{n1} is discrete and we can easily see that

$$\begin{aligned} Y_{n1} &= \begin{cases} (-1)^l \binom{k_n}{l} \left(1 - p_0^+(l)\right) - \sum_{l' \neq l} (-1)^{l'} \binom{k}{l'} p_0^+(l'), & \text{with probability } p_0^+(l) \\ -\sum_{l=1}^{k_n} (-1)^l \binom{k_n}{l} p_0^+(l), & \text{with probability } 1 - \sum_{l=1}^{k_n} p_0^+(l) \end{cases} \\ &= \begin{cases} (-1)^l \binom{k_n}{l} - Q_{k_n}, & \text{with probability } p_0^+(l) \\ -Q_{k_n}, & \text{with probability } 1 - \sum_{l=1}^{k_n} p_0^+(l). \end{cases} \end{aligned}$$

with

$$Q_{k_n} = \sum_{l=1}^{k_n} (-1)^l \binom{k_n}{l} p_0^+(l). \tag{11}$$

Thus, if we define $\sigma_{ni}^2 =: var(Z_{ni}) = var(Z_{n1})$ and $\sigma_n^2 = \sum_{i=1}^n \sigma_{ni}^2 = n\sigma_{n1}^2$ then straightforward calculations yield

$$\sigma_n^2 = \sum_{l=1}^{k_n} p_0^+(l) \left((-1)^l \binom{k_n}{l} - Q_{k_n} \right)^2 + \left(1 - \sum_{l=1}^{k_n} p_0^+(l) \right) Q_{k_n}^2. \tag{12}$$

Next, we would like to derive the Central Limit Theorem (CLT) for $\sum_{i=1}^n Z_{ni}/\sigma_n$. To be able to apply the Lindeberg CLT, we would need to obtain sharp bounds for σ_n when $k_n \to \infty$. The following auxiliary theorem, which to the best of our knowledge has not been proved before, will be needed. The result generalizes the well-known Stirling's approximation $\binom{2k}{k} \sim 4^k (\pi k)^{-1/2}$ as $k \to \infty$. Its proof exploits some of the known properties of Legendre polynomials and modified Bessel functions.

Theorem 4.1. *Define the function*

$$S_k(\alpha) = \sum_{i=0}^{k} \binom{k}{i}^2 \alpha^i$$

on $[0, 1]$ *for a given integer* $k \geq 0$. *Then for any fixed* $\alpha \in [0, 1]$, *it holds that*

$$S_k(\alpha) \sim \frac{1}{\sqrt{\pi}} \frac{(\sqrt{\alpha} + 1)^2}{\sqrt{(\sqrt{\alpha} + 1)^4 - (1 - \alpha)^2}} \frac{(\sqrt{\alpha} + 1)^{2k}}{\sqrt{k}} \tag{13}$$

as $k \to \infty$.

Now, recall Q_{k_n} from (11). For k_n such that $\lim_{n \to \infty} k_n = \infty$ we have that

$$\begin{aligned} Q_{k_n} &= \sum_{r=1}^{m} \tilde{\pi}_r (1 - \alpha_r) \sum_{l=1}^{k_n} (-1)^l \binom{k_n}{l} \alpha_r^{l-1} \\ &= \sum_{r=1}^{m} \tilde{\pi}_r \frac{1 - \alpha_r}{\alpha_r} \left\{ \sum_{l=0}^{k_n} (-1)^l \binom{k_n}{l} \alpha_r^l - 1 \right\} \end{aligned}$$

$$= \sum_{r=1}^{m} \tilde{\pi}_r \frac{1-\alpha_r}{\alpha_r} \left((1-\alpha_r)^{k_n+1} - 1\right)$$
$$\to 1 - \sum_{r=1}^{m} \frac{\tilde{\pi}_r}{\alpha_r} \quad \text{as } n \to \infty. \tag{14}$$

This together with Theorem 4.1 implies the following Corollary.

Corollary 4.1. *Let $\alpha_m \in (0,1)$ denote again the largest failure probability in p_0^+. Then, for n large enough*

$$\frac{c_m}{2} \frac{(\sqrt{\alpha_m}+1)^{2k_n}}{\sqrt{k_n}} \le \sigma_n^2 \le \frac{3c_m}{2} \frac{(\sqrt{\alpha_m}+1)^{2k_n}}{\sqrt{k_n}} \tag{15}$$

with

$$c_m = \frac{\tilde{\pi}_m(1-\alpha_m)}{\alpha_m} \frac{(\sqrt{\alpha_m}+1)^2}{\sqrt{\pi}\sqrt{(\sqrt{\alpha_m}+1)^4 - (1-\alpha_m)^2}}.$$

4.2. Asymptotic normality of the estimator of N and inference

Based on $X_1, \ldots, X_n$ i.i.d. $\sim p_0^+$, consider the empirical estimator $\bar{p}_n(j) = n^{-1}\sum_{i=1}^n \mathbb{1}_{X_i=j}$ for $j \ge 1$. The relationship in (8) is equivalent to writing

$$q_0 := -\frac{p_0(0)}{1-p_0(0)} = -(-1)^{k_n} \frac{\Delta^{k_n} p_0(0)}{1-p_0(0)} + \sum_{l=1}^{k_n} (-1)^l \binom{k_n}{l} p_0^+(l).$$

This suggests the following estimator $\widehat{q}_n$ for the true quotient q_0:

$$\widehat{q}_n = \sum_{l=1}^{k_n} (-1)^l \binom{k_n}{l} \bar{p}_n(l). \tag{16}$$

Once we derive the limit distribution of $\widehat{q}_n$ for an appropriate k_n, obtaining the limit distribution for the estimator of $1/(1-p_0(0))$ and hence for the estimator of N will be straightforward. Here, k_n can be compared to a bandwidth in kernel estimation, as it is chosen so that it fulfills a certain tradeoff between a bias and variance term (see the proof of Theorem 4.2). However, there is one main difference: Here, we are not looking for optimality. Rather, we are content with a degree of monotonicity which ensures asymptotic normality of our estimator.

Now we can state our first main result, which will be later used to establish asymptotic normality of our estimator of species richness. For this, we will need the following assumption.

Assumption A4. $\alpha_m > (\sqrt{2}-1)^2 \approx 0.171$, where α_m denotes again the largest failure probability in the finite mixture of geometric pmfs of Assumption A3.

Theorem 4.2. *Let the estimator $\widehat{q}_n$ be the one defined in (16) for the true quantity $q_0 = -p_0(0)/(1-p_0(0))$ and let the expression for σ_n be given by (12). Suppose that Assumptions A3 and A4 are satisfied. If*

$$k_n = \left\lfloor \frac{\log n}{2\log\left(\alpha_m^{1/2}+1\right)} \right\rfloor \tag{17}$$

then

$$\frac{\sqrt{n}(\widehat{q}_n - q_0)}{\sigma_n} \to_d \mathcal{N}(0,1)$$

as $n \to \infty$.

Remark 4.1. The condition that $\alpha_m > (\sqrt{2}-1)^2$ is verifiable. In fact, it can be easily shown under Assumption A3 that $\alpha_m > 1 - p_0^+(1)$. Also, it follows from the Strong Law of Large Numbers that $\bar{\alpha}_n =: 1 - \bar{p}_n(1) \to 1 - p_0^+(1)$ almost surely. Hence, with probability one, $\alpha_m > \bar{\alpha}_n$ for n large enough. This also means that if $\alpha_m \le (\sqrt{2}-1)^2$, then this would imply that $\bar{\alpha}_n < (\sqrt{2}-1)^2$ almost surely for n large enough. Therefore, provided that the sample size is large (which is the case in species richness applications) it is enough to check that $\bar{\alpha}_n \ge (\sqrt{2}-1)^2$.

Note also that the condition $\alpha_m > (\sqrt{2}-1)^2 \approx 0.171$ seems reasonable as it indicates that the mixture is well-spread, which should make estimation and inference more feasible.

Since σ_n is unknown, we would like to replace it by a consistent estimator $\widehat{\sigma}_n$, for example an estimator based on $\bar{p}_n$. Also, α_m is unknown, and hence should be substituted by a reasonable surrogate. One possibility is to use $\bar{\alpha}_n$, which is an almost sure lower bound for α_m when n is large enough.

Theorem 4.3. *Let $\bar{\alpha}_n = 1 - \bar{p}_n(1)$. Suppose that Assumptions A3 and A4 are satisfied. Also, consider the variance estimator*

$$\widehat{\sigma}_n^2 = \sum_{l=1}^{k_n} \binom{k_n}{l}^2 \bar{p}_n(l).$$

If we choose

$$k_n = \left\lfloor \frac{\log n}{2\log\left(\bar{\alpha}_n^{1/2} + 1\right)} \right\rfloor \tag{18}$$

then

$$\frac{\sqrt{n}(\widehat{q}_n - q_0)}{\widehat{\sigma}_n} \to_d \mathcal{N}(0, 1)$$

as $n \to \infty$.

The previous theorem allows us now to define our estimator of N, based on which we can construct asymptotic confidence intervals. Before doing so, we again have to adopt the notation of the species richness framework. Recall that we have switched the notation from S to n. We observe $n \sim \mathrm{Bin}(N, 1 - p_0(0))$ distinct species, and conditionally on n we observe abundances $X_1, \ldots, X_n$, which are i.i.d. $\sim p_0^+$. We denote by $\bar{p}_n$ the empirical estimator based on these abundances, $\widehat{q}_n$ the estimator of $q_0 = -\frac{p_0(0)}{1-p_0(0)}$ with an appropriate degree of monotonicity k_n. Now, define the estimator of N given by

$$\widehat{N}^\infty = \left\lfloor n(1 - \widehat{q}_n) \right\rfloor. \tag{19}$$

We are interested in establishing the limit distribution of $\widehat{N}^\infty$ for N large under the assumption that $\bar{\alpha}_n = 1 - \bar{p}_n(1) \geq (\sqrt{2} - 1)^2$. This is done in the next proposition.

Proposition 4.1. *Consider the estimator in (19). Assume that $\bar{\alpha}_n \geq (\sqrt{2} - 1)^2$ and take k_S such that it satisfies the condition in (18). Then, as $N \to \infty$*

$$\frac{1}{\sqrt{n}\widehat{\sigma}_n}\left(\widehat{N}^\infty - N\right) \to_d Z$$

with $\widehat{\sigma}_n^2 = \sum_{l=1}^{k_n} \binom{k_n}{l}^2 \bar{p}_n(l)$, and $Z \sim \mathcal{N}(0, 1)$. Thus, for any $\alpha \in (0, 1)$

$$\left[\widehat{N}^\infty - z_{1-\alpha/2}\sqrt{n}\widehat{\sigma}_n, \widehat{N}^\infty + z_{1-\alpha/2}\sqrt{n}\widehat{\sigma}_n\right]$$

is an asymptotic confidence interval with level $1 - \alpha$, with $z_{1-\alpha/2}$ denoting the $(1 - \alpha/2)$-quantile of $\mathcal{N}(0, 1)$.

5. Simulation results

In this section, we compare our approach with the ones considered by Chee and Wang (2016) (whenever it applies) and Giguelay and Huet (2018). Our basis for comparison is a simulation study where we consider 8 distributions which are either completely monotone (∞-monotone) or k-monotone for some large k ($k \geq 6$); see Table 1 where each of the 8 considered distributions was given an index from 1 to 8.

The asymptotic normality shown in Theorem 4.3 is certainly a convenient feature of the estimator $\widehat{q}_n$ which is based on the empirical estimator (of the truncated pmf). For illustration, consider the true completely monotone pmf

$$p_0(j) = 0.4 \times 0.6 \times 0.4^j + 0.6 \times 0.3 \times 0.7^j, \quad j \in \mathbb{N}$$

and the associated truncated pmf $p_0^+(j) = p_0(j)/(1 - p_0(0))\mathbb{1}_{j\geq 1}$, that is

$$p_0^+(j) = \left(\frac{8}{29} \times 0.6 \times 0.4^{j-1} + \frac{21}{29} \times 0.3 \times 0.7^{j-1}\right)\mathbb{1}_{j\geq 1}. \tag{20}$$

Note that in this example $\alpha_m = 0.7 > (\sqrt{2} - 1)^2$. For a total number of replications $M = 1000$, we generated $n = 10{,}000$ i.i.d. random variables $\sim p_0^+$, computed $\widehat{q}_n$ with $k_n = \left\lfloor \log n/(2\log(\bar{\alpha}_n^{1/2} + 1)) \right\rfloor$, $\widehat{\sigma}_n^2 = \sum_{l=1}^{k_n} \binom{k_n}{l}^2 \bar{p}_n(l)$ and the statistic $\sqrt{n}(\widehat{q}_n - q_0)/\widehat{\sigma}_n$, where $q_0 = -p_0(0)/(1-p_0(0)) = -21/29$. In Fig. 2 we plotted the histogram of the obtained realizations of the statistic, which is clearly close to the density of $\mathcal{N}(0, 1)$.

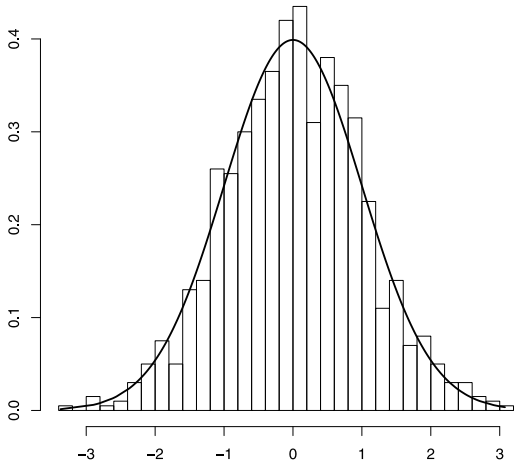

Fig. 2. Histogram of the statistic $\sqrt{n}(\widehat{q}_n - q_0)/\widehat{\sigma}_n$ based on $M = 1000$ replications where in each replication i.i.d. random variables $X_1, \ldots, X_n$, with $n = 10{,}000$, were generated from p_0^+ given in (20). The solid curve depicts the probability density function of $\mathcal{N}(0, 1)$.

Table 1
The 8 different true pmfs considered in the simulation section.

Index	$p_0(j)$	Parameters	C&W
1	$\pi(1-\alpha_1)\alpha_1^j + (1-\pi)(1-\alpha_2)\,\alpha_2^j$	$\pi = 0.3$ $(\alpha_1, \alpha_2) = (0.1, 0.8)$	$\surd$
2	$\pi(1-\alpha_1)\,\alpha_1^j + (1-\pi)(1-\alpha_2)\,\alpha_2^j$	$\pi = 0.5$ $(\alpha_1, \alpha_2) = (0.15, 0.85)$	$\surd$
3	$\pi_1(1-\alpha_1)\alpha_1^j + \pi_2(1-\alpha_2)\alpha_2^j$ $+(1-\pi_1-\pi_2)(1-\alpha_3)\alpha_3^j$	$(\pi_1, \pi_2) = (0.25, 0.5)$ $(\alpha_1, \alpha_2, \alpha_3) = (0.2, 0.4, 0.8)$	–
4	$\pi e^{-\lambda_1}\frac{\lambda_1^j}{j!} + (1-\pi)e^{-\lambda_2}\frac{\lambda_2^j}{j!}$	$\pi = 0.8$ $(\lambda_1, \lambda_2) = (0.2, 1.3)$	$\surd$
5	$\pi\,\frac{\Gamma(\gamma_1+j)}{j!\Gamma(\gamma_1)}\left(\frac{\gamma_1}{\gamma_1+\theta_1}\right)^{\gamma_1}\left(1-\frac{\gamma_1}{\gamma_1+\theta_1}\right)^j$ $+\,(1-\pi)\times\frac{\Gamma(\gamma_2+j)}{j!\Gamma(\gamma_2)}\left(\frac{\gamma_2}{\gamma_2+\theta_2}\right)^{\gamma_2}\left(1-\frac{\gamma_2}{\gamma_2+\theta_2}\right)^j$	$\pi = 0.7$ $(\gamma_1, \theta_1) = (0.5, 0.8)$ $(\gamma_2, \theta_2) = (2, 2)$	$\surd$
6	$\frac{1}{(j+1)(j+2)}$	–	–
7	$\pi e^{-\lambda}\frac{\lambda^j}{j!} + (1-\pi)Q_l^k$	$\pi = 0.3$ $(\lambda, k, l) = (0.1, 7, 5)$	–
8	$\pi Q_{l_1}^k(j) + (1-\pi)Q_{l_2}^k(j)$	$\pi = 0.5$ $(k, l_1, l_2) = (15, 4, 10)$	–

Let us comment on some of the distributions displayed in Table 1. The pmf with index 5 is a mixture of two Negative Binomial distributions with shape parameter γ_i and success probability $\pi_i = \gamma_i/(\gamma_i + \theta_i)$ for $i = 1, 2$. The pmf with index 6 is equal to the pmf obtained by drawing the parameter of a geometric distribution uniformly from [0, 1]; i.e.,

$$\frac{1}{(j+1)(j+2)} = \int_0^1 (1-\alpha)\alpha^j d\alpha.$$

This example is considered to investigate how our approach would perform for a continuous mixture of geometric distributions, thereby violating Assumption A3.

The pmf with index 8 is 15-monotone as a mixture of the 15-monotone splines Q_4^{15} and Q_{10}^{15}. Recall that Q_l^k denotes the k-monotone spline with support $\{0, \ldots, l\}$; see its expression given in (2). In Table 1 we also specify the distributions which were considered by Chee and Wang (2016) in their simulations. These are indicated by the symbol "$\surd$".

The (maximal) degree of monotonicity of the 8 distributions considered in this simulation section is given below in Table 2. The 6- and 10- monotonicity of the mixture of Poisson pmf's (with index 4) and Negative Binomial pmf's (with index 5) can be checked numerically, whereas the 7-monotonicity of the mixture of the Poisson distribution with rate $\lambda = 0.19$ and the 7-monotone spline Q_5^7 (with index 7) is an immediate consequence of the fact that the rate 0.19 lies between $\lambda_7 = 0.193$ and $\lambda_8 = 0.170$, the smallest zeros of the Laguerre polynomials L_7 and L_8 respectively. See Proposition

Table 2
Degree of monotonicity of the discrete pmf's 1-8 considered in the simulation section.

Distribution	1	2	3	4	5	6	7	8
k	∞	∞	∞	6	10	∞	7	15

2.6 of Balabdaoui et al. (2019) which gives a characterization of k-monotonicity for a Poisson distribution. See also their Table 3.

For each of the distributions 1–8 described above, the following operations were performed:

- generate $X_1, \ldots, X_N$ i.i.d. random variables $\sim p_0$ for $N = 1000$,
- remove from the sample the realizations which are identically equal to 0,
- with n, the (random) size of the remaining observations, compute the species richness estimators $\widehat{N}^{\infty}$ defined in (19), and also $\widehat{N}^{G,k_{\max}}$, the estimator proposed by Giguelay and Huet (2018). To compute the latter, we take $k_{\max} \in \{6, 10\}$, compute $\widehat{k}$ the estimate of the degree of monotonicity k, and define

$$\widehat{N}^{G,k_{\max}} = \left\lfloor n \left(1 - \sum_{l=1}^{\widehat{k}} \binom{\widehat{k}}{l} \bar{p}_n(l) \right) \right\rfloor.$$

  The estimator $\widehat{k}$ was computed using a more recent version of the R-package **pkmon**, which was kindly made available to us by François Deslandes.

Note that in Giguelay and Huet (2018) the estimator is defined without taking the integer part. Taking the integer part does not alter our conclusions but at the same time is more consistent with how the other estimators are defined. Based on 500 replications we computed for the estimators $\widehat{N}^{\infty}$ and $\widehat{N}^{G,k_{\max}}$ the following statistics

- $\bar{N}$: the empirical mean,
- $\check{N}$: the median,
- sd: the standard deviation,
- RMSE: the root mean square error,
- the asymptotic (and symmetric) 95% confidence interval.

For the distributions 1, 2, 4, and 5 we report the results of Chee and Wang (2016) given in their Table 3. Our numerical findings are given in Tables 3 and 5. Ranking the estimators according to their RMSE, we highlight in bold the statistics for the best and second best estimators. Here, we require that both the standard deviation and the RMSE of the second best estimator do not exceed those of the best one by more than 20 units. For all the settings we considered, we observe that $\widehat{N}^{\infty}$ is either the best or second best estimator. One important feature is that our estimator gives good results even when complete monotonicity is not satisfied. For example, for the distribution with index 5, which is 10-monotone, our method gives much sharper results than $\widehat{N}^{G,10}$. The 95%-confidence interval corresponding to the latter has a negative lower end for all of the 8 distributions considered. This is due to very large variance of the estimator, since in this case the selected $\widehat{k}$ turned out to be exactly equal to $k_{\max}$ in all of the 500 replications.

To obtain a fairer comparison with $\widehat{N}^{CW}$ for the distributions 1, 2, 4 and 5 considered in Chee and Wang (2016), we ran additional simulations in order to compare the length of our confidence interval with theirs. In Table 4 we give the average, median and standard deviation of the length of the 95%- confidence intervals based on our estimator $\widehat{N}^{\infty}$ by replicating 100 times the simulation steps described above (this means that for these distributions we used 50,000 replications in total). For the distributions 1, 2 and 4 we observe that the mean and median of the lengths is much smaller than the length of the 95% -confidence interval reported by the authors. Note also that the standard deviation of the lengths of our confidence intervals is relatively small. For the distribution with index 5, the length of our confidence interval is larger. However, our intervals are computed on the basis of 500 replications, whereas Chee and Wang (2016) considered only 200.

6. Testing for complete monotonicity

Before using our approach to real data, one should first test for complete monotonicity of the unknown pmf generating the abundances of the observed species. This will be done using the complete monotone LSE studied by Balabdaoui and de Fournas-Labrosse (2020). In this section, we present a nonparametric test which is be based on bootstrapping a suitable statistic.

Testing complete monotonicity of some pmf based on observed i.i.d random variables requires having at hand an estimator of this unknown pmf which is (ideally) computed under this hypothesis. Fortunately, the completely monotone LSE which was first considered by Balabdaoui and de Fournas-Labrosse (2020) can be used in this testing problem. Before describing the test statistic and the bootstrap procedure used to find the critical region, we first recall the definition of

Table 3
Simulations results for the distributions 1, 2, 4 and 5: in each row are reported the statistics corresponding to the different estimators of species richness which are compared in this paper. In the row $\widehat{N}^{CW}$ we report the statistics published by Chee and Wang (2016) in their Table 3. See text for more details.

Method	$\bar{N}$	$\check{N}$	sd	RMSE	95%-CI
	1				
$\widehat{N}^{\infty}$	**836.25**	**830.00**	**141.02**	**216.00**	**(579.55, 1109.57)**
$\widehat{N}^{G,6}$	842.96	846.00	259.90	303.44	(311.65, 1313.72)
$\widehat{N}^{G,10}$	648.932	772.50	2980.40	2998.04	(−5658.12, 6200.55)
$\widehat{N}^{CW}$	**877.21**	**832.48**	**169.51**	**208.97**	**(723.99, 1374.17)**
	2				
$\widehat{N}^{\infty}$	**800.42**	**783.50**	**113.58**	**229.58**	**(631.37, 1073.57)**
$\widehat{N}^{G,6}$	842.34	838.00	219.16	229.00	(414.55 , 1288.05)
$\widehat{N}^{G,10}$	809.70	972.00	2497.95	2502.69	(−3749.50,5890.02)
$\widehat{N}^{CW}$	990.67	832.78	631.13	629.62	(678.82, 2567.86)
	4				
$\widehat{N}^{\infty}$	**952.01**	**941.50**	**182.54**	**188.56**	**(602.70, 1309.00)**
$\widehat{N}^{G,6}$	**950.20**	**941.50**	**178.59**	**185.23**	**(598.32, 1281.00)**
$\widehat{N}^{G,10}$	1160.60	1132.00	913.04	926.16	(−607.67, 2982.10)
$\widehat{N}^{CW}$	938.00	864.66	413.89	417.48	(602.77, 1791.73)
	5				
$\widehat{N}^{\infty}$	**877.35**	**879.00**	**159.37**	**200.98**	**(580.42, 1182.10)**
$\widehat{N}^{G,6}$	892.77	902.00	267.18	287.64	(373.17, 1401.15)
$\widehat{N}^{G,10}$	1067.52	1106.50	2241.71	2240.48	(−3186.92, 5359.02)
$\widehat{N}^{CW}$	**896.16**	**853.37**	**159.30**	**189.82**	**(758.29, 1258.65)**

Table 4
Empirical mean, median and standard deviation of the length of the 95%-confidence intervals for the true N using our estimation approach when the true distribution has index $\in \{1, 2, 4, 5\}$; see also Table 1. These statistics were based on 100 replications (where for each replication, the confidence intervals were obtained on the basis of 500 runs). The 4th row in the table gives the length of the 95%-confidence interval obtained by Chee and Wang (2016) based on 200 runs.

Distribution	1	2	4	5
Mean	558.91	422.85	701.36	614.99
Median	558.63	423.07	700.30	612.30
Standard deviation	21.58	17.70	30.07	30.40
Length of the CI in Chee and Wang (2016)	650.18	1889.04	1188.96	500.36

Table 5
Simulations results for the distributions 3, 6, 7 and 8: in each row are reported the statistics corresponding to the estimators of species richness $\widehat{N}^{\infty}$, $\widehat{N}^{G,6}$ and $\widehat{N}^{G,10}$. See text for more details.

Method	$\bar{N}$	$\check{N}$	sd	RMSE	95%-CI
	3				
$\widehat{N}^{\infty}$	**925.12**	**926.00**	**139.94**	**158.59**	**(648.47, 1196.35)**
$\widehat{N}^{G,6}$	959.18	954.00	227.49	230.89	(521.42, 1378.97)
$\widehat{N}^{G,10}$	1056.93	1151.50	1923.96	1922.87	(−2919.40, 507.52)
	6				
$\widehat{N}^{\infty}$	**858.99**	**852.50**	**133.93**	**194.38**	**(594.37, 1117.00)**
$\widehat{N}^{G,6}$	885.19	874.00	229.80	256.67	(425.37, 1313.00)
$\widehat{N}^{G,10}$	1115.76	1103.50	2222.00	2222.80	(−3156.60, 5515.35)
	7				
$\widehat{N}^{\infty}$	**982.32**	**973.00**	**199.21**	**199.80**	**(604.85, 1393.00)**
$\widehat{N}^{G,6}$	980.21	973.00	**197.16**	**197.95**	**(604.85, 1386.00)**
$\widehat{N}^{G,10}$	1094.82	1078.00	924.68	928.61	(−853.87, 2992.00)
	8				
$\widehat{N}^{\infty}$	**944.41**	**952.00**	**181.52**	**189.67**	**(563.32, 1311.62)**
$\widehat{N}^{G,6}$	942.43	952.00	**177.00**	**185.96**	**(563.32, 1264.02)**
$\widehat{N}^{G,10}$	1025.92	1088.00	931.15	930.58	(−853.25, 2898.77)

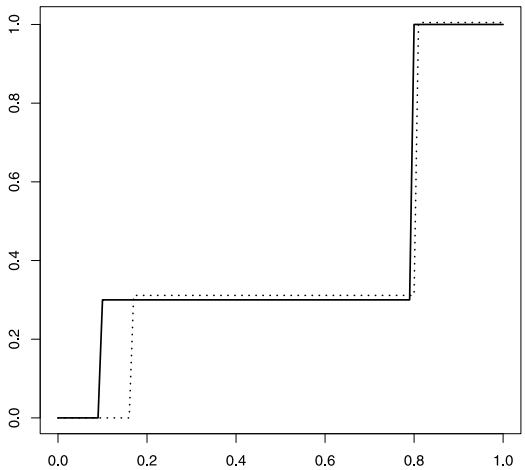

Fig. 3. The true mixing distribution $F_0(x) = 0.3\mathbb{1}_{x\geq 0.1} + 0.7\mathbb{1}_{x\geq 0.8}$ in depicted solid curve and its completely monotone LSE in points. The random sample used is of size $n = 1000$.

this estimator and some of its properties. Note that although the pmf which we are considering in the context of species richness is supported on $\mathbb{N} \setminus \{0\}$, the change of variable $j \mapsto j - 1$ re-defines complete monotonicity on $\mathbb{N}$. In the sequel, we denote by $\ell_1(\mathbb{N})$ and $\ell_2(\mathbb{N})$ the classes of functions p defined on $\mathbb{N}$ such that $\sum_{j\in\mathbb{N}} p(j) < \infty$ and $\sum_{j\in\mathbb{N}} p^2(j) < \infty$ respectively. Consider the class $\mathcal{M}$ of functions p defined on $\mathbb{N}$, which is given by

$$\mathcal{M} = \{p : p \text{ is completely monotone on } \mathbb{N}\} \cap \ell_2(\mathbb{N}).$$

It follows from Proposition 2.1 of Balabdaoui and de Fournas-Labrosse (2020) that $p \in \mathcal{M} \cap \ell_2(\mathbb{N})$ if and only if there exists a non-negative and non-decreasing function F defined on $[0, 1]$ such that $F(1-) = F(1)$ and $p(j) = \int_0^1 (1-\alpha)\alpha^j dF(j)$ for all $j \in \mathbb{N}$. Thus, when p is a pmf, we obtain from this result that p is a mixture of a geometric distributions with integral representation as in (4).

Now, given n i.i.d. random variables $X_1, \ldots, X_n \sim p_0 \in \mathcal{M}$, we define the nonparametric completely monotone LSE as

$$\begin{aligned}\widetilde{p}_n &= \operatorname{argmin}_{p\in\mathcal{M}} \|p - \bar{p}_n\|_2^2 \\ &= \operatorname{argmin}_{p\in\mathcal{M}} \sum_{j\in\mathbb{N}} (p(j) - \bar{p}_n(j))^2 \\ &= \operatorname{argmin}_{p\in\mathcal{M}} \Big\{ \sum_{j\in\mathbb{N}} p^2(j) - 2\sum_{j\in\mathbb{N}} \bar{p}_n(j)p(j) \Big\},\end{aligned}$$

where $\bar{p}_n$ denotes again the empirical estimator. Existence and uniqueness of $\widetilde{p}_n$ can be easily established. Furthermore, $\widetilde{p}_n$ is necessarily a finite mixture of geometric distributions and converges at the rate $\sqrt{n}$ in any ℓ_q-norm, provided that $q \in [2, \infty]$; see Proposition 2.3 and Proposition 3.1 in Balabdaoui and de Fournas-Labrosse (2020). Computation of $\widetilde{q}_n$ can be done efficiently using the support reduction algorithm described in Groeneboom et al. (2008). Although $\widetilde{p}_n$ is not a genuine pmf (the elements of the class $\mathcal{M}$ are not constrained to be pmfs necessarily), its total mass converges to 1. For an illustration, see Fig. 3 where we have plotted the complete monotone LSE of the mixing distribution when the true pmf is $p_0(j) = \pi(1-\alpha_1)\alpha_1^j + (1-\pi)(1-\alpha_2)\alpha_2^j,\ j \in \mathbb{N}$ for $(\pi, \alpha_1, \alpha_2) = (0.3, 0.1, 0.8)$ (which is one of the pmf's considered by Chee and Wang (2016); see also the distribution with index 1 in Table 1). The total mass of the LSE in this example is ≈ 1.005.

Consider now the following statistic:

$$T_n := \sqrt{n} \sum_{j\geq 0} |\widetilde{p}_n(j) - \bar{p}_n(j)|\widetilde{p}_n(j),$$

which estimates deviation from complete monotonicity in a weighted ℓ_1-norm. Large values of T_n should indicate violation of this shape constraint. Although the limit distribution of T_n was derived by Balabdaoui and de Fournas-Labrosse (2020) in their Corollary 3.11, this limit depends unfortunately on the unknown pmf in a complicated way. Hence, we resort to a bootstrap procedure to test

$$H_0 : p_0 \text{ is completely monotone} \quad \text{versus} \quad H_1 : p_0 \text{ is not.}$$

This bootstrap procedure can be described as follows: given a large integer $B > 0$, and for $b = 1, \ldots, B$

Table 6
The estimated power (in %) of the bootstrap test based on random samples from one of the pmfs 1–11 and with size $n \in \{100, 500\}$. Estimation is based on $M = 100$ replications. For each replication the conditional quantile is estimated based on a bootstrap sample of size $B = 100$. The star symbol "$\star$" indicates that the estimated power was found to be 100%.

n	1	2	3	4	5	6	7	8	9	10	11
100	5	2	2	2	4	0	2	3	45	83	$\star$
500	8	6	1	1	5	1	9	4	96	$\star$	$\star$

Table 7
Specification of the species in the 4 considered datasets, and values of $\bar{\alpha}_n = 1 - \bar{p}_n(1)$, where $\bar{p}_n$ is the empirical estimator of the abundances of the observed species.

Dataset	Data I	Data II	Data III	Data IV
Species	Claims	Worms	Words sung	Words known
$\bar{\alpha}_n$	0.187	0.617	0.496	0.533

- draw a random sample $(X_1^{(b)}, \dots, X_n^{(b)})$ from the completely monotone pmf
$$\frac{\widetilde{p}_n}{\sum_{j\in\mathbb{N}} \widetilde{p}_n(j)},$$
- based on the sample $(X_1^{(b)}, \dots, X_n^{(b)})$ compute the bootstrap empirical and completely monotone estimators $\bar{p}_n^{(b)}$ and $\widetilde{p}_n^{(b)}$ respectively, and the corresponding bootstrap statistic
$$T_n^{(b)} = \sqrt{n} \sum_{j\geq 0} |\widetilde{p}_n^{(b)}(j) - \bar{p}_n^{(b)}(j)| \widetilde{p}_n^{(b)}(j).$$

Let $\widetilde{q}_{n,B,1-\alpha}$ be the empirical $(1-\alpha)$-quantile of $(\widetilde{T}_n^{(1)}, \dots, \widetilde{T}_n^{(B)})$. Then, H_0 is rejected at the level α if $T_n > \widetilde{q}_{n,B,1-\alpha}$.

To investigate numerically the power of the proposed test, we consider in addition to the distributions 1–8 given above in Table 1 the following pmfs (to which we assign indices 9, 10 and 11):

9. $p_0(j) = 0.5^j e^{-0.5}/j!,\ j \in \mathbb{N}$, which is only convex (2-monotone) on $\mathbb{N}$,

10. $p_0(j) = e^{-1}/j!, j \in \mathbb{N}$, which is only monotone decreasing (1-monotone) on $\mathbb{N}$,

11. $p_0(j) = \binom{50}{j} 2^{-50} \mathbb{1}_{j\in\{0,\dots,50\}}$, which is unimodal with mode at $j = 25$ (and hence is not k-monotone for any $k \in \mathbb{N}$).

The results of the bootstrap procedure described above are given in Table 6 for $n \in \{100, 500\}$. The bootstrap quantile was estimated on the basis of $B = 100$ runs. Here $\alpha = 0.05$. $M = 100$ replications were used to estimate the power of our test under the 11 distributions described above.

The test has a good power for the alternative distributions considered (unimodal and not monotone, 1- and 2-monotone) even for moderate sample size $n = 100$. Conservativeness of the test is observed for the pmfs under H_0, represented by 1,2, 3 and 6. This phenomenon is somehow expected and is known to happen for tests in other shape-constrained problems such as estimation under convexity; see e.g. Balabdaoui et al. (2017) and Balabdaoui et al. (2018). Note that for the pmfs with indices 4, 5, 7 and 8, which are 6-, 10-, 7- and 15-monotone respectively (and hence almost completely monotone), the test has low power. This is rather expected and even desired. In fact, we would like that our method could be used when k-monotonicity holds with large k as the currently available k-monotonicity tests have difficulties in deciding between k and $k+1$ in such cases; we refer again to Section 3.2 in Giguelay and Huet (2018).

7. Application to real data

In this section, we show how our method can be applied to real data. We consider 4 different datasets. Table 7 gives a quick overview of what "species" means in the context of each of these datasets.

For the 4 datasets the bootstrap test described in the previous section led to the acceptance of the complete monotonicity hypothesis. Fig. 4 shows the empirical and completely monotone estimators of the abundances in Data I, II and III. The same plot for Data IV was already shown above in Fig. 1. Also, the condition $\bar{\alpha}_n = 1 - \bar{p}_n(1) \geq (\sqrt{2}-1)^2$ is valid for all these datasets as shown in Table 7. Since complete monotonicity is a valid model, the selected degree of monotonicity $\widehat{k}$ as defined by Giguelay and Huet (2018) will take the value of the chosen k_{max}. This phenomenon has been observed in the results of the simulations described above , with $k_{max} \in \{6, 10\}$, and in other simulations with $k_{\max} = 20$ whose results are not shown here. This gives non-sensible results for large values k_{max} due to the huge variance of $\widehat{N}^{G,k_{\max}}$ in this case. For this reason, we compare only our estimator $\widehat{N}^{\infty}$ to the one obtained by Chee and Wang (2016) for Data I, II and IV. This indirectly allows for a further comparison with other estimators reported by the authors. These include the MLE of a mixture of Poisson of Norris and Pollock (1998) and the convex LSE of Durot et al. (2015).

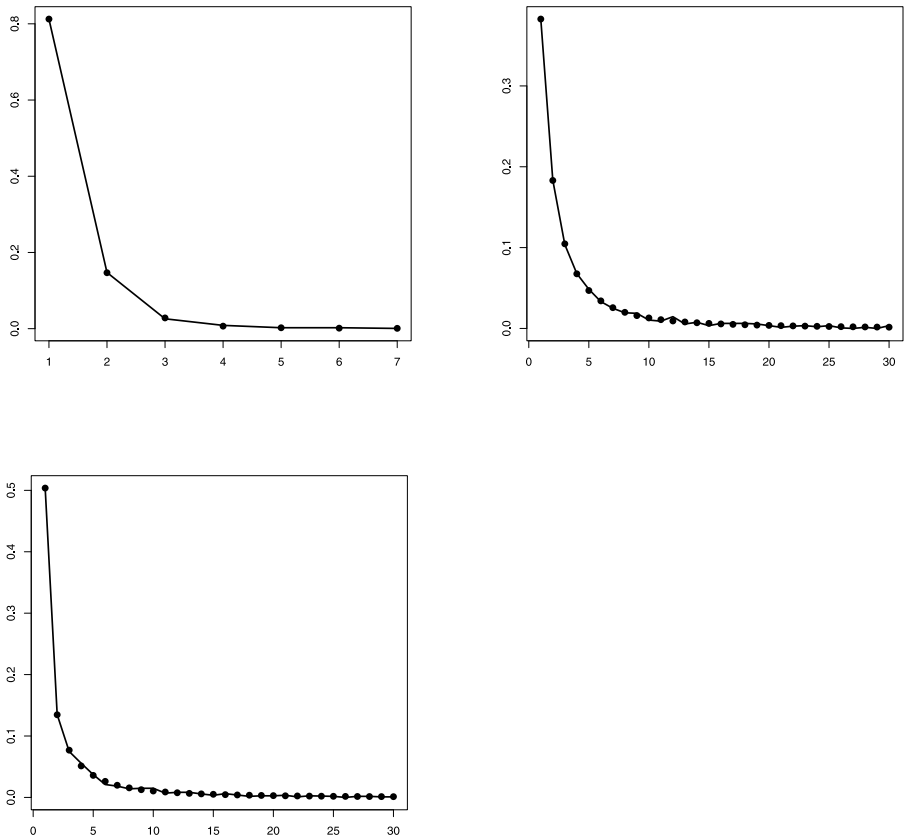

Fig. 4. In solid line is depicted the empirical estimator of the abundances for the species in the insurance claim, Filarial worms and Robbie Williams data (left, right, bottom). In bullets is the completely monotone LSE.

Table 8
Estimation results for the insurance claims data using the completely monotone approach described here and the one considered by Chee and Wang (2016).

True N	$\widehat{N}^{CW}$	$\widehat{N}^{\infty}$	Completely monotone 95%-CI
9461	7374	6424	(3462.77, 9385.22)

7.1. Data I: Insurance claims

The data consists of accident insurance policies which reported a certain (positive) number of claims during a given year. The number of claims reported ranges from 1 to 7. In this dataset, we do not have access to the original insurance policy data, but only to their empirical frequencies. If n_j denotes the number of policies which reported j claims for $j \in \{1, \ldots, 7\}$, then it was observed that $n_1 = 1317$, $n_2 = 239$, $n_3 = 42$, $n_4 = 14$, $n_5 = 4$, $n_6 = 4$ and $n_7 = 1$. Thus, the number of observed claims is $n = \sum_{i=j}^{7} n_j = 1621$. The total number of policies for this data is known and equal to $N = 9461$.

As shown in Table 8, we obtain $\widehat{N}^{\infty} = 6424$, whereas the estimate of Chee and Wang (2016) is $\widehat{N}^{CW} = 7374$, obtained by fitting the unconditional discretized k-monotone MLE with $k = 9$. Hence both approaches seem to underestimate the truth in this case. This can occur even though complete monotonicity fits very well this dataset. The completely monotone 95%-confidence interval is as given in Proposition 4.1. This confidence interval does not contain N, which is consistent with the aforementioned underestimation.

7.2. Data II: Filarial worms

The filarial worms dataset contains the observed frequencies of these worms in rats and mice and was kindly provided by Yong Wang upon request. The data can be found in Baayen (2001) and have appeared earlier in Bertram (1949) in a study of the transmission mechanisms of the worms. The true N is known here as well and is equal to 2600. The results are shown in Table 9. The estimate $\widehat{N}^{CW} = 2787$ was found using the unconditional discretized k-monotone MLE with $k = 19$, while our estimate $\widehat{N}^{\infty} = 2689$ is closer to the true N. Also, the latter belongs to our 95%-confidence interval.

Table 9
Estimation results for the filarial worms data using the completely monotone approach described here and the one considered by Chee and Wang (2016).

True N	$\widehat{N}^{CW}$	$\widehat{N}^{\infty}$	Complete monotone 95%-CI
2600	2787	2689	(1894.91, 3483.09)

Table 10
Estimation results for the Shakespeare data using the completely monotone approach described here and the one considered by Chee and Wang (2016).

True N	$\widehat{N}^{CW}$	$\widehat{N}^{\infty}$	Completely monotone 95%-CI
–	112 605	78 171	(61 626.08, 94 715.92)

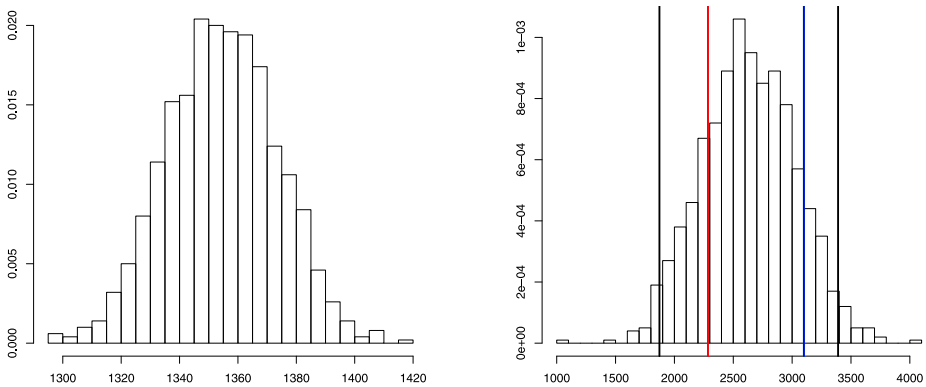

Fig. 5. Left: Histogram of S, the number of words found in the independent sub-samples drawn from the complete dataset of Robbie Williams. Right: Histogram of the estimator $\widehat{N}^{\infty}$ based on S. The red and blue vertical lines depict respectively the values 2285 (the total number of unique words sung) and 3100, the number of words counted by the lyrics website "musixmatch". The black vertical lines show the 2.5% and 97.5% empirical quantiles. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

7.3. Data III: Number of words sung by Robbie Williams

The dataset of Robbie Williams's lyrics was created by scanning about 40 songs found in his albums. The data was generously shared by Gregor Bachmann, a student at ETH Zurich, who is the creator of these data. Here, species are represented by the unique words sung by Robbie Williams. The whole dataset contains 17 438 words of which only 2285 are distinct. In this example, we will view $N = 2285$ as the true total number of words sung by the singer with the goal of estimating it. From the original sample, we drew with replacement 1000 independent sub-samples of size 8000. Based on each sub-sample, we computed the number of distinct words S and the corresponding estimator of the words richness, $\widehat{N}^{\infty}$. Histograms of both S and $\widehat{N}^{\infty}$ are shown in Fig. 5. The mean value and standard deviation of $\widehat{N}^{\infty}$ based on this sampling were found to be 2631.66 and 395.40 respectively. As shown in the histogram of $\widehat{N}^{\infty}$, the 95% -confidence interval contains the total number of unique words sung. Interestingly, the same interval also contains the value 3100, which is the number of words counted for the Pop star by the website "musixmatch", known for being the first website for lyrics.

7.4. Data IV: Vocabulary richness of Shakespeare

How many words did Shakespeare know? It is an interesting question that has preoccupied a number of statisticians; see e.g. Efron and Thisted (1976). One can of course safely assume that this great author knew most of the English words that one can find in a dictionary. A dataset of words used by Shakespeare can be found in Spevack (1968). Chee and Wang (2016) noticed a discrepancy between the counts used by Efron and Thisted (1976) and those given in Spevack (1968). They recomputed the word counts up to 100 occurrences directly from Appendix of Spevack (1968); see Table 5 in Chee and Wang (2016). To apply our method we used the same counts given in that table, which contains $195\,008 = \sum_{j=1}^{100} jn_j$ words (and not 884 650 words as reported by Chee and Wang (2016)), among which 30 792 are distinct.

Our estimation method gives $\widehat{N}^{\infty} = 78\,171$, which is significantly below the estimate obtained (Chee and Wang, 2016), $\widehat{N}^{CW} = 112\,605$. Note that $\widehat{N}^{CW}$ is outside our 95%-confidence interval. Our estimate is larger than the one based on Poisson mixtures ($\sim 72\,500$ words), while the method by Durot et al. (2015) gives a much smaller estimate (of order 56 000) (see Table 10).

8. Some conclusions

In this paper we proposed a nonparametric estimation method of species richness in a given population under the assumption that the distribution of species abundances is completely monotone, or equivalently is a mixture of geometric pmfs. Our interest in this class stems from the fact that complete monotonicity is the limit of k-monotonicity as $k \to \infty$, and hence can be used as a natural surrogate for the k-monotone model when k is large. To simplify the analysis, our focus was put on the sub-class of finite mixtures. Our main result shows that as the sample size $n \to \infty$, the k_n-monotone pmf, where k_n is a function of $\log n$, is an adequate approximation of the true geometric mixture that guarantees asymptotic normality of the estimator for species richness. As opposed to the selection approach suggested by Giguelay and Huet (2018) where a maximum range for k has to be chosen in advance, or the Akaike-type criterion used by Chee and Wang (2016), our method is fully automatic and can be quickly implemented since it only requires quantities that can be immediately computed using the empirical estimator. Also, this logarithmic dependence on n guards against fitting very large k which may result in aberrant estimates with very large variance.

Our method seems to work well even when the true distribution is only k-monotone for a finite and relatively large k; e.g., $k \geq 6$. This is a very desirable feature which we find to be coherent with the fact that our complete monotonicity test (based on the LSE of Balabdaoui and de Fournas-Labrosse (2020)) has a low power for such distributions. When compared with the methods proposed by Chee and Wang (2016) and Giguelay and Huet (2018), our approach to estimating species richness gives very often better results with smaller variances and narrower confidence intervals.

Another possible approach to estimating species richness under complete monotonicity is maximum likelihood estimation. Some simple facts on the complete monotone MLE such as existence, uniqueness and finiteness of the associated mixing distribution can be shown using the seminal work of Lindsay (1983). More challenging is establishing its rate of convergence and limit distribution. We conjecture that, like the completely monotone LSE, the MLE converges at $\sqrt{n}$. A proof would require tools from the empirical process theory. We intend to investigate this problem in a separate work.

Acknowledgments

The first author would like to thank professor Emmanuel Kowalski for some discussions about the proof of Theorem 4.1.

Appendix

A.1. Proofs for results of Section 2

Proof of Proposition 2.1. Let $k \in \mathbb{N}$ be large enough. Define

$$F^k(\alpha) = \sum_{0 \leq j \leq k\alpha} \binom{k}{j} (-1)^{k-1-j} \Delta^{k-1-j} p(j)$$

for all $\alpha \in [0, 1)$, and the corresponding completely monotone non-negative sequence

$$p^k(i) = \int_0^1 (1-\alpha)\alpha^i dF^k(\alpha) = \sum_{j=0}^{k-1} \binom{k}{j} \left(\frac{j}{k}\right)^i \left(1 - \frac{j}{k}\right) (-1)^{k-1-j} \Delta^{k-1-j} p(j) \tag{21}$$

for all $i \in \mathbb{N}$. We show first that $\lim_{k\to\infty} p^k(i) = p(i)$ for all $i \in \mathbb{N}$. Note that for $r, i \in \mathbb{N}$ we have that

$$(-1)^r \Delta^r p(i) = \int_0^1 \alpha^i (1-\alpha)^{r+1} dF(\alpha).$$

From plugging this identity in (21) and writing $u_i(\alpha) = \alpha^i(1-\alpha)$ it follows that

$$\begin{aligned} p^k(i) &= \sum_{j=0}^{k-1} u_i(j/k) \int_0^1 \binom{k}{j} \alpha^j (1-\alpha)^{k-j} dF(\alpha) \\ &= \int_0^1 \sum_{j=0}^{k} u_i(j/k) \binom{k}{j} \alpha^j (1-\alpha)^{k-j} dF(\alpha), \quad \text{since } u_i(1) = 0 \\ &= \int_0^1 B_{k,u_i}(\alpha) dF(\alpha), \end{aligned}$$

where, using the same notation as in Feller (1971), $B_{k,u}$ denotes the Bernstein polynomial of degree k and corresponding to a given function u. Thus, the preceding display is equal to $E[B_{k,u_i}]$, the expectation of B_{k,u_i} with respect to F. Using the fact that $B_{k,u}$ converges uniformly to u on $[0, 1]$, it follows that for any $i \in \mathbb{N}$

$$\lim_{k\to\infty} p^k(i) = p(i).$$

To show that F^k converges weakly to F, note first that $F^k(\alpha) \in [0, 1]$ since

$$F^k(1-) = \int_0^1 \sum_{j=0}^{k-1} \binom{k}{j} \alpha^j (1-\alpha)^{k-j} dF(\alpha) = \int_0^1 (1-\alpha^k) dF(\alpha) \leq 1$$

using that $\sum_{j=0}^{k} \binom{k}{j} \alpha^j (1-\alpha)^{k-j} = 1$. It follows from the Helly–Bray theorem (see for example Lemma 2.5 in van der Vaart (1998)) that there exists a subsequence k' along which $F^{k'}$ converges weakly to a sub-distribution on $[0, 1]$, $\tilde{F}$ say; i.e., $F^{k'}$ converges vaguely to $\tilde{F}$. Since u_i is a continuous and bounded function on $[0, 1]$, it follows from the definition of weak convergence that

$$\lim_{k' \to \infty} p^{k'}(i) = \int_0^1 u_i(\alpha) d\tilde{F}(\alpha) = \int_0^1 \alpha^i (1-\alpha) d\tilde{F}(\alpha).$$

Since $p^{k'}(i)$ has to converge to $p(i)$ we should have $\int_0^1 \alpha^i (1-\alpha) dF(\alpha) = \int_0^1 \alpha^i (1-\alpha) d\tilde{F}(\alpha)$ for all $i \in \mathbb{N}$. Uniqueness of the mixing measure in the completely monotone representation implies that $F = \tilde{F}$, and the proof is complete. □

Proof of Proposition 2.2. For all $k, i \in \mathbb{N}$ we have that $p^k(i) \in [0, 1]$. Using again the diagonalization technique of the Helly–Bray Theorem we can show that there exists a subsequence k' along which $p^{k'}(i)$ converges to some limit $p(i)$ for any $i \in \mathbb{N}$ as $k' \to \infty$. To keep the notation light we shall write k for k'. We have that

$$\begin{aligned}\int_0^1 \alpha^i (1-\alpha) dF^k(\alpha) &= \sum_{j=0}^{k-1} \left(\frac{j}{k}\right)^i \left(1 - \frac{j}{k}\right) \pi_j^{k-1-j} \frac{k}{k-j} \\ &= \sum_{j=0}^{k-1} \binom{k-1}{j} \left(\frac{j}{k}\right)^i (-1)^{k-1-j} \Delta^{k-1-j} p^{k-1-j}(j).\end{aligned}$$

Using the general reciprocity formula (1.8) in Feller (1971) with $\nu = k-1$, $c_j = p_j^{k-1-j}$ and $a_{i'+j} = ((i'+j)/k)^i$ with $i' = 0$ (here i' is used instead of i in formula (1.8)), we get that

$$\begin{aligned}\int_0^1 \alpha^i (1-\alpha) dF^k(\alpha) &= \sum_{l=0}^{k-1} p^{k-1-l}(l) \binom{k-1}{l} \sum_{j=0}^{l} \binom{l}{j} (-1)^{l-j} \left(\frac{j}{k}\right)^i \\ &= p^{k-1-i}(i) \binom{k-1}{i} \frac{i!}{k^i} \\ &\quad + \sum_{l=0}^{i-1} p^{k-1-l}(l) \binom{k-1}{l} \sum_{j=0}^{l} \binom{l}{j} (-1)^{l-j} \left(\frac{j}{k}\right)^i,\end{aligned}$$

where the preceding equality is a consequence of the following identity:

$$\sum_{j=0}^{l} \binom{l}{j} (-1)^j j^i = (-1)^l l! S(i, l),$$

where $S(i, l)$ denotes the Stirling number of the second kind, known to satisfy $S(i, l) = 0$ for all $l > i$ and $S(i, i) = 1$. Besides $\lim_{k\to\infty} \binom{k-1}{i} \frac{i!}{k^i} = 1$, and $\lim_{k\to\infty} \binom{k-1}{l} \frac{1}{k^i} = 0$, so we can deduce the following equality:

$$p(i) = \lim_{k\to\infty} \int_0^1 \alpha^i (1-\alpha) dF^k(\alpha).$$

By assumption the sequence F^k converges weakly to a distribution function F such that $F(1-) = 1$. Thus the limit on the right of the previous display is also equal to $\int_0^1 \alpha^i (1-\alpha) dF(\alpha)$, which in turn implies that p is the completely monotone pmf with F as mixing distribution. Now recall that p was the pointwise limit of a subsequence $p^{k'}$. Uniqueness of p implies that the original sequence, p^k, should converge pointwise to p, as $k \to \infty$. This completes the proof of the proposition. □

A.2. Proofs for results of Section 4

Proof of Theorem 4.1. First recall that the Legendre polynomial of degree k can be given by the following expression

$$L_k(x) = \frac{1}{2^k} \sum_{i=0}^{k} \binom{k}{i}^2 (x-1)^i (x+1)^{k-i}$$

$$= \frac{(x+1)^k}{2^k} \sum_{i=0}^{k} \binom{k}{i}^2 \left(\frac{x-1}{x+1}\right)^i.$$

Putting $\alpha = (x-1)/(x+1)$ yields the identity

$$S_k(\alpha) = (1-\alpha)^k L_k\left(\frac{\alpha+1}{1-\alpha}\right).$$

It follows from 14.15.13 in Olver et al. (2010) that uniformly in $\xi \in (0, \infty)$

$$L_k(\cosh\xi) = \sqrt{\frac{\xi}{\sinh\xi}}\, I_0((k+1/2)\xi)\left(1 + O\left(\frac{1}{k}\right)\right)$$

where $z \mapsto I_\nu(z)$ is the modified Bessel function, the solution of the differential equation

$$z^2 w''(z) + z w'(z) - (z^2 + \nu^2) w(z) = 0.$$

Now, it follows from 9.7.1 in Abramowitz and Stegun (1992) that for ν fixed, $z \in \mathbb{R}$ large and $\mu = 4\mu^2$

$$I_\nu(z) \sim \frac{e^z}{\sqrt{2\pi z}}\left(1 + \frac{1}{8z} + o\left(\frac{1}{z}\right)\right). \tag{22}$$

Let ξ such that $\cosh\xi = (\alpha+1)/(1-\alpha)$. Elementary calculus gives that

$$\begin{aligned}\xi &= \log\left(\frac{\alpha+1}{1-\alpha} + \sqrt{\left(\frac{\alpha+1}{1-\alpha}\right)^2 - 1}\right)\\ &= \log\left(\frac{(\sqrt{\alpha}+1)^2}{1-\alpha}\right)\end{aligned}$$

and

$$\begin{aligned}\frac{e^\xi/2}{\sqrt{\sinh\xi}} &= \frac{e^\xi/2}{\sqrt{(e^\xi - e^{-\xi})/2}}\\ &= \frac{\sqrt{2}e^\xi}{e^{2\xi}-1}\\ &= \frac{\sqrt{2}(\sqrt{\alpha}+1)^2}{\sqrt{(\sqrt{\alpha}+1)^4 - (1-\alpha)^2}}.\end{aligned}$$

Also, it follows from (22) that as $k \to \infty$

$$I_0((k+1/2)\xi) \sim \frac{e^{\xi/2}}{\sqrt{2\pi}} \frac{e^{k\xi}}{\sqrt{k}}\left(1 + O\left(\frac{1}{k}\right)\right).$$

Thus,

$$\begin{aligned}S_k(\alpha) &\sim \frac{1}{\sqrt{\pi}} \frac{(\sqrt{\alpha}+1)^2}{\sqrt{(\sqrt{\alpha}+1)^4 - (1-\alpha)^2}} \frac{e^{k[\log(1-\alpha) + \log((\sqrt{\alpha}+1)^2/(1-\alpha))]}}{\sqrt{k}}\\ &= \frac{1}{\sqrt{\pi}} \frac{(\sqrt{\alpha}+1)^2}{\sqrt{(\sqrt{\alpha}+1)^4 - (1-\alpha)^2}} \frac{e^{2k\log(\sqrt{\alpha}+1)}}{\sqrt{k}}\\ &= \frac{1}{\sqrt{\pi}} \frac{(\sqrt{\alpha}+1)^2}{\sqrt{(\sqrt{\alpha}+1)^4 - (1-\alpha)^2}} \frac{(\sqrt{\alpha}+1)^{2k}}{\sqrt{k}}\end{aligned}$$

which is the claimed result.

Proof of Corollary 4.1. We have that

$$\begin{aligned}1 - \sum_{l=1}^{k_n} p_0^+(l) &= 1 - \sum_{r=1}^{m} \tilde{\pi}_r (1-\alpha_r) \sum_{l=1}^{k_n} \alpha_r^{l-1}\\ &= 1 - \sum_{r=1}^{m} \tilde{\pi}_r (1 - \alpha_r^{k_n})\end{aligned}$$

$$= \sum_{r=1}^{m} \tilde{\pi}_r \alpha_r^{k_n} \to 0, \quad \text{as } n \to \infty.$$

Rewriting the expression for σ_n^2 given in (12) we get

$$\begin{aligned}
\sigma_n^2 &= \sum_{l=1}^{k_n} p_0^+(l) \left((-1)^l \binom{k_n}{l} - Q_{k_n} \right)^2 + \left(1 - \sum_{l=1}^{k_n} p_0^+(l) \right) Q_{k_n}^2 \\
&= \sum_{l=1}^{k_n} p_0^+(l) \binom{k_n}{l}^2 - 2Q_{k_n} \sum_{l=1}^{k_n} (-1)^l \binom{k_n}{l} p_0^+(l) + Q_{k_n}^2 \\
&= \sum_{l=1}^{k_n} p_0^+(l) \binom{k_n}{l}^2 - Q_{k_n}^2, \quad \text{by definition of } Q_{k_n} \qquad (23) \\
&= \sum_{r=1}^{m} \frac{\tilde{\pi}_r (1-\alpha_r)}{\alpha_r} \left(S_{k_n}(\alpha_r) - 1 \right) - Q_{k_n}^2 \\
&= \sum_{r=1}^{m} \frac{\tilde{\pi}_r (1-\alpha_r)}{\alpha_r} S_{k_n}(\alpha_r) + 1 - \sum_{r=1}^{m} \frac{\tilde{\pi}_r}{\alpha_r} - Q_{k_n}^2 \qquad (24)
\end{aligned}$$

where $S_k(\alpha)$ is from Theorem 4.1. By the same theorem, and the proved convergence in (14), the bounds in (15) follow. □

Proof of Theorem 4.2. We first note that

$$\frac{\sqrt{n}(\widehat{q}_n - q_0)}{\sigma_n} = \frac{\sum_{i=1}^n Z_{ni}}{\sigma_n} + \frac{(-1)_n^k \Delta^{k_n} p_0(0)}{1 - p_0(0)} \frac{\sqrt{n}}{\sigma_n}$$

We shall handle the random and deterministic parts separately. Since $\sigma_{ni}^2/\sigma_n^2 = 1/n$, the random sequence $\sum_{i=1}^n Z_{ni}/\sigma_n$ converges weakly to $\mathcal{N}(0, 1)$ if and only if

$$\lim_{n\to 0} \frac{1}{\sigma_n^2} \sum_{i=1}^n E\left[Z_{ni}^2 \mathbb{1}_{|Z_{ni}| \geq \epsilon \sigma_n} \right] = 0$$

for any fixed $\epsilon > 0$. The latter condition is also equivalent to

$$\lim_{n\to 0} \frac{1}{\sigma_n^2} E\left[Y_{n1}^2 \mathbb{1}_{|Y_{n1}| \geq \sqrt{n}\epsilon\sigma_n} \right] = 0 \qquad (25)$$

where we recall that $Y_{n1} = \sqrt{n} Z_{n1}$. The condition in (25) is satisfied if

$$\frac{|Y_{n1}|}{\sqrt{n}\sigma_n} \to_{a.s.} 0. \qquad (26)$$

Since $|Y_{n1}|$ takes the $k_n + 1$ values $|(-1)^l \binom{k_n}{l} - Q_{k_n}|, l = 1, \ldots, k_n$ and $|Q_{k_n}|$, the almost sure convergence in (26) is equivalent to

$$\lim_{n\to\infty} \frac{\left\{ \max_{1 \leq l \leq k_n} \left| (-1)^l \binom{k_n}{l} - Q_{k_n} \right| \right\} \vee |Q_{k_n}|}{\sqrt{n}\sigma_n} = 0.$$

Without loss of generality, we take k_n to be even. Since Q_n converges to a finite limit; see (14), the latter means that

$$\lim_{n\to\infty} \frac{\binom{k_n}{k_n/2}}{\sqrt{n}\sigma_n} = 0. \qquad (27)$$

Using the well-known asymptotic Stirling's approximation $\binom{k_n}{k_n/2} \sim 2^{k_n}/\sqrt{k_n}$ and the lower bound for σ_n^2 from (15), we get that for n large enough

$$\begin{aligned}
\frac{\binom{k_n}{k_n/2}}{\sqrt{n}\sigma_n} &\asymp \frac{2^{k_n}(\alpha_m^{1/2} + 1)^{-k_n}}{\sqrt{n} k_n^{1/4}} \\
&= \exp\left\{ k_n \log\left(\frac{2}{\alpha_m^{1/2} + 1} \right) - \frac{\log n}{2} - \frac{1}{4} \log(k_n) \right\}.
\end{aligned}$$

Thus, convergence in (27) follows if

$$k_n \le \frac{\log n}{2\log\left(\frac{2}{\alpha_m^{1/2}+1}\right)}, \quad \text{and} \quad \lim_{n\to\infty} k_n = \infty. \tag{28}$$

The inequality in (28) holds true (and is even strict) if k_n satisfies (17) given in the statement of the theorem. Indeed, $\alpha_m > (\sqrt{2}-1)^2$ implies that $\alpha_m^{1/2}+1 > 2/(\alpha_m^{1/2}+1)$ and hence

$$\frac{1}{\log(\alpha_m^{1/2}+1)} < \frac{1}{\log\left(\frac{2}{\alpha_m^{1/2}+1}\right)}.$$

Also, we obviously have that $\lim_{n\to\infty} k_n = \infty$, which implies that $-1/4\log(k_n) \to -\infty$.

We now need to study the bias term. As $n \to \infty$ we have that

$$\begin{aligned}\frac{\sqrt{n}(-1)^{k_n}\Delta^{k_n}p_0(0)}{\sigma_n} &\asymp \sqrt{n}\frac{(1-\alpha_1)^{k_n}}{(\alpha_m^{1/2}+1)^{k_n}}k_n^{1/4} \\ &= \exp\left\{\frac{\log n}{2} - k_n\log\left(\frac{\alpha_m^{1/2}+1}{1-\alpha_1}\right) + \frac{1}{4}\log(k_n)\right\}.\end{aligned} \tag{29}$$

When k_n satisfies (17), then

$$k_n \ge \frac{\log n}{2\log(\alpha_m^{1/2}+1)} - 1$$

and hence

$$\begin{aligned}\frac{\log n}{2} - k_n\log\left(\frac{\alpha_m^{1/2}+1}{1-\alpha_1}\right) &= \frac{\log n}{2\log(\alpha_m^{1/2}+1)}\log(1-\alpha_1) + \log\left(\frac{\alpha_m^{1/2}+1}{1-\alpha_1}\right) \\ &\to -\infty, \quad \text{as } n\to\infty\end{aligned}$$

since $\log(1-\alpha_1) < 0$. We conclude that the term in (29) must also diverge to $-\infty$ since the term $\log(k_n)$ is of a lower order. This in turn implies that

$$\lim_{n\to\infty}\frac{\sqrt{n}(-1)^{k_n}\Delta^{k_n}p_0(0)}{(1-p_0(0))\sigma_n} = 0.$$

This proves the claimed asymptotic normality of the estimator $\widehat{q}_n$. □

Remark A.1. The sufficient condition in (26) used above to ensure the Lindeberg CLT condition is essentially necessary in the following sense: the convergence has to happen in probability, and hence almost surely for some subsequence. Indeed, it follows from the Markov inequality that

$$P(Y_{n1}^2 > n\sigma_n^2\epsilon^2) \le \frac{E(Y_{n1}^2)}{n\sigma_n^2\epsilon^2} = \frac{1}{n\epsilon^2}$$

since $\sigma_n^2 = E(Y_n^2)$.

Proof of Theorem 4.3. From Remark 4.1 we know that with probability 1, there exists $N_0 > 1$ such that for all $n \ge N_0$ we have $\alpha_m > \bar{\alpha}_n$. In the following, we will omit writing "almost surely". We will now check that the convergence in (27) occurs when k_n satisfies the new condition (18). We have that

$$\begin{aligned}k_n\log\left(\frac{2}{\alpha_n^{1/2}+1}\right) - \frac{\log n}{2} &\le \frac{\log n}{2\log(\bar{\alpha}_n^{1/2}+1)}\log\left(\frac{2}{\alpha_m^{1/2}+1}\right) - \frac{\log n}{2} \\ &= \frac{\log n}{2\log(\bar{\alpha}_n^{1/2}+1)}\log\left(\frac{2}{(\bar{\alpha}_n^{1/2}+1)(\alpha_m^{1/2}+1)}\right) \\ &< 0\end{aligned}$$

since $(\bar{\alpha}_n^{1/2}+1)(\alpha_m^{1/2}+1) > (\bar{\alpha}_n^{1/2}+1)^2 \ge 2$, which follows from the condition $\alpha_n \ge (\sqrt{2}-1)^2$.

Also, when k_n satisfies (18) we have that

$$k_n > \frac{\log n}{2\log(\alpha_m^{1/2}+1)} - 1.$$

Hence, using the same inequality as in the proof of Theorem 4.2 we conclude again that the bias term converges to 0.

Next, we show that $\widehat{\sigma}_n/\sigma_n \to_p 1$ as $n \to \infty$., We can write

$$\begin{aligned}\frac{\widehat{\sigma}_n^2}{\sigma_n^2} - 1 &= \frac{\sum_{l=1}^{k_n}\binom{k_n}{j}^2\left(\bar{p}_n(l) - p_0^+(l)\right)}{\sigma_n^2} + \frac{\sum_{l=1}^{k_n}\binom{k_n}{j}^2 p_0^+(l) - \sigma_n^2}{\sigma_n^2}\\ &= T_{n1} + T_{n2}\end{aligned}$$

where by the expression obtained in (23)

$$T_{n2} = \frac{Q_{k_n}^2}{\sigma_n^2} \to 0 \quad \text{as } n \to \infty.$$

Now, we turn to T_{n1} which can be rewritten as

$$T_{n1} = \frac{1}{n\sigma_n^2}\sum_{i=1}^{n} W_{ni}$$

with $W_{ni} = \sum_{l=1}^{k_n}\binom{k_n}{l}^2\left(\mathbb{1}_{X_i=l} - p_0^+(l)\right)$. Put $v_n^2 = \sum_{l=1}^{k_n} p_0^+(l)\binom{k_n}{l}^2$. Then,

$$\begin{aligned}Var(W_{n1}) &= \sum_{l=1}^{k_n} p_0^+(l)\left(\binom{k_n}{l}^2 - v_n^2\right)^2 + \left(1 - \sum_{l=1}^{k_n} p_0^+(l)\right)v_n^4\\ &= \sum_{l=1}^{k_n} p_0^+(l)\binom{k_n}{l}^4 - v_n^4 \le \left(\sum_{l=1}^{k_n}\sqrt{p_0^+(l)}\binom{k_n}{l}^2\right)^2\end{aligned}$$

using the inequality $\sqrt{x+y} \le \sqrt{x} + \sqrt{y}$ which holds for any $x, y \ge 0$. The same inequality can be applied to get

$$\begin{aligned}\sum_{l=1}^{k_n}\sqrt{p_0^+(l)}\binom{k_n}{l}^2 &\le \sum_{r=1}^{m}\tilde{\pi}_r^{1/2}\frac{(1-\alpha_r)^{1/2}}{\alpha_r^{1/2}}\sum_{l=1}^{k_n}\binom{k_n}{l}^2\alpha_r^{l/2}\\ &\asymp \frac{(\alpha_m^{1/4}+1)^{2k_n}}{\sqrt{k_n}}\end{aligned}$$

using Theorem 4.1. It follows that

$$\begin{aligned}Var(T_{n1}) &\asymp \frac{(\alpha_m^{1/4}+1)^{4k_n}}{n\sigma_n^4 k_n}\\ &\asymp \frac{(\alpha_m^{1/4}+1)^{4k_n}}{nk_n}\frac{k_n}{(\alpha_m^{1/2}+1)^{4k_n}}\\ &= \exp\left\{4k_n\log\left(\frac{\alpha_m^{1/4}+1}{\alpha_m^{1/2}+1}\right) - \log n\right\}.\end{aligned}$$

Hence, a sufficient condition for $Var(T_{n1}) \to 0$ is that the argument of the exponential diverges to $-\infty$. Straightforward calculations give

$$4k_n\log\left(\frac{\alpha_m^{1/4}+1}{\alpha_m^{1/2}+1}\right) - \log n \le \frac{\log n}{\log(\bar{\alpha}_n^{1/2}+1)}\log\left(\frac{(\alpha_m^{1/4}+1)^2}{(\alpha_m^{1/2}+1)^2(\bar{\alpha}_n+1)}\right).$$

Define the function $g(t) = (t+1)/(t^2+1)$ for $t \in [0, 1]$. Analysis of the variations of g reveals that g is unimodal with a unique maximum at $\sqrt{2}-1$. Thus, g is decreasing on $[(\sqrt{2}-1)^{1/2}, 1]$, implying that

$$\frac{(\alpha_m^{1/4}+1)^2}{(\alpha_m^{1/2}+1)^2} < \frac{(\bar{\alpha}_n^{1/4}+1)^2}{(\bar{\alpha}_n^{1/2}+1)^2}.$$

Finally, consider the function $h(t) = (t+1)^2/(t^2+1)^3$ for $t \in [(\sqrt{2}-1)^{1/2}, 1]$. This function can be shown to be decreasing on this interval with $h((\sqrt{2}-1)^{1/2}) \approx 0.96 < 1$. We conclude that

$$\log\left(\frac{(\alpha_m^{1/4}+1)^2}{(\alpha_m^{1/2}+1)^2(\bar{\alpha}_n+1)}\right) < 0,$$

which implies that $\lim_{n\to\infty} Var(T_{n1}) = 0$. The proof of the claimed asymptotic normality is complete by applying Slutsky's Theorem. □

Proof of Proposition 4.1. Recall that n is the number of species. Since

$$n(1-\widehat{q}_n)-1<\widehat{N}^{\infty}\leq n(1-\widehat{q}_n)$$

it is enough to establish the weak convergence for $s(1-\widehat{q}_s)$. By Theorem 4.3, it follows that conditionally on n

$$\frac{\sqrt{n}\,(1-\widehat{q}_n-(1-q_0))}{\widehat{\sigma}_n}|n\to_d\mathcal{N}(0,1)$$

which implies that

$$\frac{\sqrt{n}\,(1-\widehat{q}_n-(1-q_0))}{\widehat{\sigma}_n}\to_d\mathcal{N}(0,1)$$

since the limit distribution does not depend on n. Also, since $1-q_0=(1-p_0(0))^{-1}$ we have that

$$\begin{aligned}\frac{1}{\sqrt{n}\widehat{\sigma}_n}\Big(n(1-\widehat{q}_n)-N\Big)&=\frac{\sqrt{n}\,(1-\widehat{q}_n-(1-q_0))}{\widehat{\sigma}_n}+\frac{1}{\sqrt{n}\widehat{\sigma}_n}\left(\frac{n}{1-p_0(0)}-N\right)\\&=\frac{\sqrt{n}\,(1-\widehat{q}_n-(1-q_0))}{\widehat{\sigma}_n}\\&\quad+\frac{1}{\widehat{\sigma}_n}\frac{\sqrt{n}}{\sqrt{N}}\sqrt{N}\left(\frac{1}{1-p_0(0)}-\frac{N}{n}\right)\end{aligned}\tag{30}$$

where by the Central Limit Theorem and δ-method

$$\sqrt{N}\left(\frac{1}{1-p_0(0)}-\frac{N}{n}\right)\to_d\frac{1}{(1-p_0(0))^2}\mathcal{N}(0,1).$$

Furthermore, $\sqrt{n}/\sqrt{N}\to_{a.s.}\sqrt{1-p_0(0)}$ by the SLLN and continuous mapping Theorem, and $\widehat{\sigma}_n\to_p\infty$ as $N\to\infty$ (and hence $n\to\infty$ almost surely). Thus, the second term in (30) converges to 0 in probability, which yields the claimed asymptotic normality. The statement about the asymptotic confidence interval is straightforward. □

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